Unraveling Temporal and Structural Drivers of Stablecoin Issuance: A SHAP-Enhanced Deep Forecasting Framework / 解构稳定币发行量的时变与结构性驱动因素：基于SHAP增强的深度预测框架

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Abstract

Stablecoins serve as the bridge between the crypto ecosystem and the traditional financial system, and their aggregate issuance (market capitalization) is widely viewed as a key barometer of crypto-market liquidity and risk appetite. Forecasting stablecoin issuance, however, faces three challenges: the high-dimensional noise induced by a large array of on-chain, macro-financial, and sentiment variables; the black-box nature of deep learning models that limits their economic interpretability; and the neglect of structural differences between centralized collateralized and decentralized algorithmic stablecoins.

稳定币是加密生态与传统金融体系之间的桥梁，其总发行量（市值）被广泛视为加密市场流动性与风险偏好的关键晴雨表。然而，稳定币发行量预测面临三大挑战：链上、宏观金融与情绪变量众多所引致的高维噪声；深度学习模型的黑箱性质限制了其经济可解释性；以及现有研究对中心化抵押型与去中心化算法型稳定币之间结构性差异的忽视。

We propose a SHAP-enhanced deep forecasting framework for stablecoin issuance, disciplined end-to-end by a nested walk-forward validation design, with three core contributions. First, we reposition SHAP from a post-hoc explanation tool to an ex-ante feature selector (SHAP-FS) and evaluate it within a nested walk-forward cross-validation design. Second, using daily data on USDT, USDC, and DAI over 2020–2026, we show that the random walk—the benchmark that defeats every individual machine-learning model in our experiment—can itself be beaten once forecasts are *anchored, combined, and disciplined*: a horizon-pooled non-negative least-squares ensemble that averages three feature-selection strategies, keeps random-walk anchors as candidate columns, and then applies validation walk-forward shrinkage with amplitude clipping, directional blending, and evidence-based gating outperforms the no-drift random walk on RMSE, MAE, and directional accuracy at

all four horizons, with a positive out-of-sample R^2 throughout (across all 48 tasks: RMSE -6.4%, MAE -8.8%, directional accuracy 63.3% versus 40.1% for the random walk, mean $R^2_{OS} = 12.2\%$, mean DM statistic -3.2). Third, we construct a two-dimensional attribution framework—temporal (1-day, 1-week, 1-month, 3-month) and structural (centralized vs. decentralized)—to map the heterogeneity of issuance drivers across 12 prediction tasks.

本文提出了一个面向稳定币发行量预测的SHAP增强深度预测框架，并以嵌套前向交叉验证设计严格约束，做出三项核心贡献。第一，我们将SHAP从事后解释工具重新定位为事前特征选择器（SHAP-FS），并在嵌套前向交叉验证设计中对其进行评估。第二，利用2020–2026年USDT、USDC与DAI的日度数据，我们证明：随机游走——这一在本实验中击败了每一个单独机器学习模型的基准——本身可以通过锚定、组合与约束被击败。一个按期限池化的非负最小二乘集成，对三种特征选择策略取等权平均、保留随机游走锚作为候选列，再施以验证集前向收缩、幅度截断、方向混合与证据门控，在全部四个期限上同时在RMSE、MAE与方向准确性上超越无漂移随机游走，且样本外R^2始终为正（全部48个任务上：RMSE降低6.4%、MAE降低8.8%，方向准确性63.3%对随机游走的40.1%，平均$R^2_{OS} = 12.2\%$，平均DM统计量-3.2）。第三，我们构建了二维归因框架——时间维度（1天、1周、1月、3月）与结构维度（中心化vs去中心化）——在12个预测任务上刻画发行驱动因素的异质性。

The empirical design follows a three-stage pipeline: data preparation and model construction, feature selection and forecast evaluation, and driver heterogeneity analysis. Starting from an initial pool of **104** candidate predictors spanning macro-financial, Fed-liquidity-plumbing, crypto-global, derivatives-leverage, exchange-flow, depeg-arbitrage, on-chain-structural, and autoregressive categories, the SHAP-FS procedure selects a sparse subset for each task through a five-phase nested process. All explanations are based on out-of-sample SHAP values. Two findings organize the results. On the one hand, no *individual* model beats the no-drift random walk: LSTM, Transformer, random forest, and XGBoost all record higher out-of-sample RMSE than the RW benchmark, a stablecoin analogue of the Meese–Rogoff puzzle. On the other hand, once the individual forecasts are pooled by horizon, averaged across the three feature-selection strategies, stacked with non-negative weights that explicitly include the random walk as an anchor column, and disciplined by validation walk-forward shrinkage with amplitude clipping, directional blending, and evidence-based gating, the resulting SHAP-FS Ensemble beats the random walk on every horizon and every evaluation metric, and the improvement is robust to the shrinkage multiplier over a wide band. The sparse SHAP-FS representation remains central to the framework: it matches the all-feature ensemble's accuracy with about seven predictors per task while revealing that short-horizon issuance is dominated

by autoregressive supply momentum and market-wide realized-cap flows, long-horizon issuance is anchored in Federal Reserve liquidity-plumbing variables (ON RRP and the Treasury General Account) alongside short-term interest rates through the reserve-yield and money-market channel, and the decentralized stablecoin DAI depends more on coin-specific on-chain quantities than its centralized peers.

实证设计遵循三阶段流程：数据准备与模型构建、特征选择与预测评估、驱动因素异质性分析。我们从涵盖宏观金融、美联储流动性管道、加密全局、衍生品杠杆、交易所资金流、脱锚套利、链上结构及自回归类别的**104**个初始候选预测变量出发，通过五阶段嵌套流程为每个任务选择稀疏子集。所有解释均基于样本外SHAP值。两组发现统领全文结果。一方面，没有任何单个模型能击败无漂移随机游走：LSTM、Transformer、随机森林与XGBoost的样本外RMSE均高于RW基准，构成稳定币版的Meese–Rogoff难题。另一方面，一旦将各模型预测按期限池化、对三种特征选择策略取等权平均、以显式包含随机游走锚的非负权重堆叠，并以验证集前向收缩、幅度截断、方向混合与证据门控加以约束，所得SHAP-FS集成在每个期限、每项评估指标上都击败随机游走，且该改进在宽幅收缩倍率区间内稳健。稀疏的SHAP-FS表示仍是框架核心：它以每项任务约七个预测变量达到与全特征集成相当的精度，同时揭示出短期发行由自回归供给动量与市场层面已实现市值资金流主导、长期发行通过储备收益与货币市场渠道锚定于美联储流动性管道变量（ON RRP与财政部一般账户TGA）及短期利率、去中心化稳定币DAI较中心化同类更依赖币种特定的链上变量。

The framework offers actionable implications for regulators, investors, and issuers, providing a data-driven feature-selection protocol for risk early warning and liquidity management.

该框架为监管机构、投资者与发行方提供了可操作的政策与实务启示，并为风险预警和流动性管理提供了数据驱动的特征筛选协议。

Keywords: stablecoin issuance forecasting, SHAP feature selection, deep learning, LSTM, Transformer, walk-forward validation, driver heterogeneity, financial machine learning 稳定币发行量预测, SHAP特征选择, 深度学习, LSTM, Transformer, 前向验证, 驱动因素异质性, 金融机器学习

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1. Introduction / 引言

Stablecoins have become a cornerstone of the cryptocurrency ecosystem, providing a stable-value bridge between the volatile crypto market and the traditional financial system. Unlike other cryptocurrencies, a stablecoin aims to maintain a stable value relative to a fiat currency—typically the US dollar—which makes it an essential vehicle for trading, remittances, decentralized-finance (DeFi) collateral, and store-of-value in the crypto space (Dionysopoulos and Urquhart, 2024; Arner et al., 2020; Eichengreen, 2019). The total issuance of stablecoins, measured by their market capitalization, is therefore widely regarded as a key indicator of liquidity conditions and risk appetite in the broader crypto market (Griffin and Shams, 2020; Olk and Miebs, 2025). Understanding and forecasting stablecoin issuance matters not only for market participants but also for regulators, because issuance dynamics can serve as an early signal of dollar-liquidity spillovers and cross-border capital flows (Gorton et al., 2025; Lyons and Viswanath-Natraj, 2023).

稳定币已成为加密货币生态系统的基石，在波动的加密市场与传统金融体系之间提供了稳定的价值桥梁。与其它加密货币不同，稳定币旨在维持相对于法定货币（通常是美元）的稳定价值，因而成为加密领域交易、汇款、去中心化金融（DeFi）抵押品以及价值存储的重要载体 (Dionysopoulos and Urquhart, 2024; Arner et al., 2020; Eichengreen, 2019)。稳定币总发行量（以市值衡量）因此被广泛视为更广泛加密市场流动性状况与风险偏好的关键指标 (Griffin and Shams, 2020; Olk and Miebs, 2025)。理解并预测稳定币发行量不仅对市场参与者重要，对监管者也同样重要，因为发行动态可充当美元流动性外溢与跨境资本流动的先行信号 (Gorton et al., 2025; Lyons and Viswanath-Natraj, 2023)。

Forecasting stablecoin issuance nevertheless presents several unique challenges. First, the crypto market is extremely data-rich, offering a vast array of on-chain metrics, derivatives-leverage indicators, macroeconomic variables, and market-sentiment indices. This high-dimensional feature space is often dominated by redundant and noisy signals, making it difficult to identify the true drivers of issuance dynamics (Guyon and Elisseeff, 2003). Second, although deep learning models have proved powerful in capturing nonlinear and time-varying patterns in financial time series (Sezer et al., 2020; Lim and Zohren, 2021; Fischer and Krauss, 2018), they are frequently criticized for their lack of economic interpretability; stakeholders need to understand not only *what* will happen but also *why* it will happen. Third, existing studies tend to treat all stablecoins as a homogeneous group, overlooking the structural differences between centralized collateralized stablecoins (e.g., USDT, USDC) and decentralized algorithmic stablecoins (e.g., DAI) (Catalini and de Gortari, 2021; Li and Mayer, 2021); these differences in issuance mechanisms suggest that their circulations may be driven by systematically different factors.

然而，稳定币发行量预测面临几个独特的挑战。首先，加密市场数据极其丰富，包含大量链上指标、衍生品杠杆指标、宏观经济变量与市场情绪指数。这种高维特征空间往往被冗余和噪声信号主导，使得难以识别发行动态的真正驱动因素 (Guyon and Elisseeff, 2003)。其次，尽管深度学习模型在捕捉金融时间序列的非线性与时变模式方

面已被证明十分强大 (Sezer et al., 2020; Lim and Zohren, 2021; Fischer and Krauss, 2018)，但其缺乏经济可解释性而常受诟病；利益相关者不仅需要知道会发生什么，还需要知道为什么会发生。第三，现有研究往往将所有稳定币视为同质群体，忽视了中心化抵押型稳定币（如USDT、USDC）与去中心化算法型稳定币（如DAI）之间的结构性差异 (Catalini and de Gortari, 2021; Li and Mayer, 2021)；这些发行机制的差异表明，其流通量可能由系统性不同的因素驱动。

To address these challenges, this paper develops a SHAP-enhanced deep forecasting framework for stablecoin issuance, coupling deep learning with SHAP (SHapley Additive exPlanations) (Lundberg and Lee, 2017) under a nested walk-forward validation design. Machine learning paired with SHAP-based interpretation has only recently reached stablecoin markets (Liu and Shen, 2025), and its use for *issuance* forecasting remains unexplored. We make three contributions to the forecasting literature. First, we reposition SHAP from a post-hoc explanation tool to an ex-ante feature selector (SHAP-FS) and evaluate whether the global feature-importance ranking it produces can be exploited *before* final model training to select informative subsets. Second, we resolve—rather than merely document—a stablecoin analogue of the Meese–Rogoff puzzle (Meese and Rogoff, 1983). In our nested walk-forward experiment, every individual machine-learning model loses to the no-drift random walk in out-of-sample RMSE, echoing the classic exchange-rate result; however, drawing on the forecast-combination literature (Roccazzella et al., 2022; Qian et al., 2022; Dai et al., 2022), we show that a horizon-pooled non-negative least-squares ensemble that averages three feature-selection strategies, retains the random walk as an explicit anchor, and is disciplined by validation walk-forward shrinkage, amplitude clipping, directional blending, and evidence-based gating beats the random walk on RMSE, MAE, and directional accuracy at all four horizons and delivers a strictly positive out-of-sample R^2 everywhere (mean 12.2%, versus deeply negative values for every single-strategy ensemble at the three-month horizon). Forecasts of stablecoin issuance therefore contain exploitable information, but harvesting it requires anchoring and regularization rather than any single sophisticated learner. Third, we construct a two-dimensional attribution framework—horizons (1-day, 1-week, 1-month, 3-month) crossed with stablecoin types (centralized vs. decentralized)—that systematically maps the structural heterogeneity of issuance drivers across 12 prediction tasks (3 stablecoins $\times$ 4 horizons).

为应对这些挑战，本文开发了一个用于稳定币发行量预测的SHAP增强深度预测框架，在嵌套前向交叉验证设计下将深度学习与SHAP (Lundberg and Lee, 2017)相结合。机器学习与SHAP解释的组合直到近期才进入稳定币市场研究 (Liu and Shen, 2025)，而将其用于发行量预测尚属空白。我们对预测文献做出三项贡献。第一，我们将SHAP从

事后解释工具重新定位为事前特征选择器（SHAP-FS），并评估其产生的全局特征重要性排序能否在最终模型训练之前被利用来选择信息子集。第二，我们不仅记录而且解决了一个稳定币版的Meese–Rogoff难题 (Meese and Rogoff, 1983)。在我们的嵌套前向实验中，每个单独的机器学习模型在样本外RMSE上都输给无漂移随机游走，呼应了经典汇率结果；然而，借助预测组合文献 (Roccazzella et al., 2022; Qian et al., 2022; Dai et al., 2022)，我们证明：一个按期限池化、显式保留随机游走锚的非负最小二乘集成，辅以朝向无变化预测的保守收缩，在全部四个期限的RMSE、MAE与方向准确性上击败随机游走，并处处取得严格为正的样本外R^2。因此，稳定币发行量预测中蕴含可利用的信息，但获取它需要的是锚定与正则化，而非任何单一精巧的学习器。第三，我们构建了二维归因框架——期限（1天、1周、1月、3月）× 稳定币类型（中心化vs去中心化）——在12个预测任务（3种稳定币×4个期限）上系统刻画发行驱动因素的结构性异质性。

The study is motivated by four research gaps identified in the literature. We summarize these gaps and our response to each in Table 1.

本研究的动机源于文献中识别出的四个研究缺口。表 1 概括了这些缺口以及本文的回应。

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 outlines the three-stage research framework. Section 4 describes the data and methodology, including the SHAP-FS procedure and the nested walk-forward validation design. Section 5 presents the empirical results. Section 6 reports the heterogeneity analysis. Section 7 discusses the findings, and Section 8 concludes.

本文其余部分组织如下：第 2 节回顾相关文献；第 3 节概述三阶段研究框架；第 4 节描述数据与方法论，包括SHAP-FS流程与嵌套前向交叉验证设计；第 5 节呈现实证结果；第 6 节报告异质性分析；第 7 节讨论研究发现；第 8 节总结全文。

2. Literature Review / 文献综述

The relevant literature spans four strands: the economics of stablecoins, the determinants of stablecoin growth, machine-learning and deep-learning approaches to cryptocurrency forecasting, and explainable artificial intelligence (XAI) with feature selection in finance. We review each strand in turn and then position this paper.

相关文献横跨四个分支：稳定币的经济学、稳定币增长的驱动因素、加密货币预测中的机器学习与深度学习方法，以及金融领域的可解释人工智能（XAI）与特征选择。下面逐一回顾，并在此基础上定位本文。

表 1: Four research gaps and the positioning of this study | 四大研究缺口与本文定位

Gap	What the literature shows	What remains unresolved
G1: Benchmark robustness	In financial time series, the random walk is notoriously difficult to beat (Meese and Rogoff, 1983; Welch and Goyal, 2008); stablecoin issuance is strongly trending and autocorrelated, so naive extrapolation already provides a strong benchmark.	Whether *any* disciplined framework can beat the random walk on stablecoin issuance out of sample, and which design choices (anchoring, combination, shrinkage) make that possible.
G2: Predictor selection	Single-theory predictors (e.g., interest rates or on-chain activity alone) are too narrow; data-rich on-chain plus macro sets introduce redundancy and noise.	Feature selection that preserves nonlinear signals while avoiding structural overfitting.
G3: Interpretability link	ML/DL flexibility is well established, but black-box forecasts limit regulatory and risk-management use.	SHAP explanations should be validated *ex ante* for prediction, not merely reported *ex post*.
G4: Driver heterogeneity	Most stablecoin studies focus on a single coin (typically USDT) or a single horizon.	A systematic mapping of drivers across horizons and across issuance mechanisms (centralized/decentralized).

缺口	文献已有发现	尚未解决的问题
G1：基准稳健性	在金融时间序列中，随机游走 notoriously 难以被击败 (Meese and Rogoff, 1983; Welch and Goyal, 2008)；稳定币发行量具有强趋势性与自相关性，因此朴素外推已构成强基准。	是否有任何受约束的框架能在稳定币发行量样本外击败随机游走，以及哪些设计选择（锚定、组合、收缩）使这种超越成为可能。
G2：预测变量选择	单一理论预测变量（如仅利率或仅链上活动）过于狭窄；丰富的链上数据加宏观集合又引入冗余与噪声。	如何在保留非线性信号的同时避免结构性过拟合的特征选择。
G3：可解释性链接	机器学习/深度学习的灵活性已确立，但黑箱预测限制了监管与风险管理用途。	SHAP解释应在预测前被验证，而非仅在事后报告。
G4：驱动因素异质性	多数稳定币研究聚焦于单一币种（通常是USDT）或单一期限。	如何系统刻画驱动因素在不同期限与发行机制（中心化/去中心化）之间的差异。

2.1. The Economics of Stablecoins / 稳定币的经济学

Stablecoins are a new form of private money that attempts to resolve the tension between the volatility of cryptoassets and the stability required for a medium of exchange (Eichengreen, 2019; Arner et al., 2020; Catalini and de Gortari, 2021). Arner et al. (2020) document the market development of existing stablecoins (e.g., Tether, USD Coin, and Dai) and the regulatory challenges posed by prospective global stablecoins, advocating a balanced and proportional regulatory approach. Conceptually, stablecoins can be viewed through the lens of money and credit: Olk and Miebs (2025) argue that the crypto sphere has evolved into a credit-based shadow banking system in which stablecoin issuers are functionally equivalent to shadow banks, connecting the crypto sphere to the conventional banking system. Li and Mayer (2021) develop a dynamic model in which a stablecoin issuer transforms risky reserve assets into tokens of stable value and show that the system is bimodal—stability can persist for a long time, but once the peg "breaks the buck," volatility persists.

稳定币是一种新型的私人货币，试图调和加密资产的高波动性与交易媒介所需的稳定性之间的矛盾 (Eichengreen, 2019; Arner et al., 2020; Catalini and de Gortari, 2021)。Arner等(2020)记录了现有稳定币（如Tether、USD Coin和Dai）的市场发展以及潜在全球稳定币带来的监管挑战，主张采取均衡、比例式的监管路径。从概念上看，稳定币可以通过货币与信用理论加以理解：Olk和Miebs (2025)认为加密领域已演变为一个基于信用的影子银行体系，其中稳定币发行人在功能上等价于影子银行，将加密领域与传统银行体系连接起来。Li和Mayer (2021)构建了一个动态模型，其中稳定币发行人将风险储备资产转化为价值稳定的代币，并证明该体系是双峰的——稳定可以长期持续，但一旦锚定"破局"，波动便会持续。

More recently, Gorton et al. (2025) provide a model and empirical evidence on a novel source of stablecoin demand: stablecoin holders are indirectly compensated for run risk by lending their coins to crypto speculators, which links crypto speculation to the traditional financial markets where stablecoins invest their reserves. This literature establishes that stablecoin issuance is not an isolated on-chain phenomenon but is intrinsically connected to leverage cycles, reserve-asset yields, and the broader monetary system (Gorton et al., 2025; Li and Mayer, 2021; Olk and Miebs, 2025).

近期，Gorton等(2025)为稳定币需求的一个新来源提供了模型与经验证据：稳定币持有者通过将其币借给加密投机者而间接获得对挤兑风险的补偿，这使加密投机与传统金融市场（稳定币在其中投资储备）相互关联。上述文献确立了稳定币发行并非孤立的链上现象，而与杠杆周期、储备资产收益率以及更广泛的货币体系内在相连 (Gorton et al., 2025; Li and Mayer, 2021; Olk and Miebs, 2025)。

2.2. Determinants of Stablecoin Growth / 稳定币增长的驱动因素

A growing empirical literature studies what drives stablecoin issuance and supply growth. Dionysopoulos and Urquhart (2024) survey the first ten years of stablecoins, their impact on the cryptocurrency ecosystem, and open research questions. In a series of papers, Dionysopoulos, Marra, and Urquhart (2024a; 2024b; 2025) identify issuer profitability from reserve investments as the dominant driver of stablecoin growth, using Tether as the central case; their findings suggest that macro-financial conditions—especially short-term interest rates—matter for issuance through the reserve-yield channel.

关于稳定币发行与供应增长驱动因素的实证文献正在不断增长。Dionysopoulos和Urquhart (2024)综述了稳定币的前十年、其对加密货币生态系统的影响及开放的研究问题。在一系列论文中，Dionysopoulos、Marra和Urquhart (2024a; 2024b; 2025)以Tether为核心案例，识别出发行人储备投资盈利性是稳定币增长的主导驱动因素；其发现表明，宏观金融状况——尤其是短期利率——通过储备收益渠道影响发行。

A second strand focuses on the peg and arbitrage dynamics. Griffin and Shams (2020) examine whether Tether issuance influenced Bitcoin prices during the 2017 boom and find patterns consistent with supply-based manipulation. Krištoufek (2021) revisits this debate using a directed-spillover framework and finds that stablecoin issuance is mostly a reaction to, rather than a driver of, cryptoasset price movements. Ante et al. (2021) likewise find that stablecoin issuances exert limited influence on cryptocurrency returns. Lyons and Viswanath-Natraj (2023) show that peg-sustaining arbitrage is the main stabilizing force of Tether, with issuance (the supply-side analogue of central-bank intervention) playing a limited role; they also document that premiums (discounts) in the peg arise from safe-haven demand (liquidity and collateral concerns). Taken together, this literature suggests that both macro-financial conditions and high-frequency on-chain and market-microstructure signals jointly shape stablecoin issuance, yet no study to date has provided an out-of-sample forecasting framework that jointly handles high-dimensional predictors and driver heterogeneity.

第二条文献主线聚焦于锚定与套利动态。Griffin和Shams (2020)考察了2017年繁荣期间Tether发行是否影响了比特币价格，并发现与供给驱动操纵一致的证据。Krištoufek (2021)使用有向溢出框架重新审视该争论，发现稳定币发行主要是对加密资产价格运动的反应而非驱动。Ante等(2021)同样发现稳定币发行对加密货币收益率的影响有限。Lyons和Viswanath-Natraj (2023)证明维持锚定的套利是Tether的主要稳定力量，而发行（供给侧的央行干预类比）作用有限；他们还发现锚定中的溢价（折价）源于避险需求（流动性与抵押品担忧）。总体而言，该文献表明宏观金融状况与高频链上及市场微观结构信号共同塑造稳定币发行，但迄今尚无研究提供能够同时处理高维预测变量与驱

动异质性的样本外预测框架。

2.3. Machine Learning for Cryptocurrency Forecasting / 加密货币预测中的机器学习

Machine learning has been widely applied to cryptocurrency forecasting. Early work demonstrates that recurrent neural networks and LSTM can beat linear benchmarks such as ARIMA for Bitcoin direction and price (McNally et al., 2018). Subsequent studies refine this result: Jaquart et al. (2021) show that gradient-boosted trees and neural networks deliver profitable short-term Bitcoin trading signals, while Sebastião and Godinho (2021) find that machine-learning forecasts of cryptocurrencies are informative and tradable, although performance is highly sensitive to market regimes. Hybrid deep architectures that parallelize convolutional and recurrent blocks further improve volatility-type forecasts in financial markets (Choi and Shin, 2022), and functional data-analysis approaches have recently been brought to intraday cryptocurrency return forecasting (Jasiak and Zhong, 2026). At the asset-pricing level, Liu et al. (2022) document a strong three-factor structure in cryptocurrency returns, and Makarov and Schoar (2020) characterize the arbitrage frictions across crypto exchanges; machine learning has also been used to predict which cryptocurrency exchanges will remain active (Milunovich and Lee, 2022). However, the vast majority of this literature focuses on Bitcoin and other volatile cryptoassets; systematic out-of-sample forecasting of stablecoin *issuance*—itself a stock of "digital cash"—remains largely unexplored.

机器学习已被广泛应用于加密货币预测。早期工作证明，递归神经网络与LSTM能够在比特币方向与价格预测上击败ARIMA等线性基准 (McNally et al., 2018)。后续研究进一步精炼了这一结论：Jaquart等(2021)证明梯度提升树与神经网络能够产生盈利的短期比特币交易信号；Sebastião和Godinho (2021)发现加密货币的机器学习预测具有信息含量且可交易，尽管其表现对市场体制高度敏感。将卷积与递归模块并行化的混合深度架构进一步改善了金融市场中波动率类预测 (Choi and Shin, 2022)，函数型数据分析方法最近也被引入加密货币收益的日内预测 (Jasiak and Zhong, 2026)。在资产定价层面，Liu等(2022)记录了加密货币收益中稳健的三因子结构；Makarov和Schoar (2020)刻画了各加密交易所之间的套利摩擦；机器学习还被用于预测哪些加密货币交易所将保持活跃 (Milunovich and Lee, 2022)。然而，该文献绝大多数聚焦于比特币及其它高波动加密资产；对稳定币发行量（其本身是一种"数字现金"存量）的系统性样本外预测仍基本空白。

2.4. Deep Sequence Models in Financial Forecasting / 金融预测中的深度序列模型

The architecture of our core model builds on two families of deep sequence models. The long short-term memory (LSTM) network (Hochreiter and Schmidhuber, 1997) introduces gated memory cells that learn long-term dependencies, and has become a standard tool in financial forecasting (Fischer and Krauss, 2018; Sezer et al., 2020). The Transformer (Vaswani et al., 2017) replaces recurrence with self-attention, allowing the model to attend to global dependencies directly and to train in parallel; Transformer-based architectures have shown strong performance in time-series forecasting (Lim and Zohren, 2021). For financial time series, deep sequence models have been shown to outperform shallow nonlinear benchmarks in many settings, although their advantage over strong statistical baselines remains an active question (Sezer et al., 2020; Lim and Zohren, 2021). Our design follows the literature by adopting LSTM as the core forecaster and the Transformer encoder as a deep-sequence benchmark, alongside tree-based models and a no-drift random walk.

核心模型的架构建立在两类深度序列模型之上。长短期记忆（LSTM）网络 (Hochreiter and Schmidhuber, 1997)引入带门的记忆单元以学习长期依赖，已成为金融预测的标准工具 (Fischer and Krauss, 2018; Sezer et al., 2020)。Transformer (Vaswani et al., 2017)以自注意力替代递归，使模型能够直接关注全局依赖并实现并行训练；基于Transformer的架构在时间序列预测中表现出色 (Lim and Zohren, 2021)。对于金融时间序列，许多情境下深度序列模型已被证明优于浅层非线性基准，但其相对于强统计基准的优势仍是一个活跃问题 (Sezer et al., 2020; Lim and Zohren, 2021)。本文的设计沿袭文献做法：以LSTM为核心预测器，以Transformer编码器作为深度序列基准，并辅以树模型与无漂移随机游走。

2.5. Feature Selection and the Random-Walk Benchmark / 特征选择与随机游走基准

Feature selection is central to high-dimensional forecasting. Guyon and Elisseeff (2003) provide the classical treatment of variable and feature selection, emphasizing that the choice of relevance criterion should be tied to the learning task. Among penalized approaches, the lasso (Tibshirani, 1996) imposes an ℓ_1 penalty and shrinks coefficients to zero, yielding a sparse linear feature-selection benchmark. In finance, Welch and Goyal (2008) and Rapach et al. (2010) show that individual predictors are unstable out of sample and that combining information across predictors improves equity-premium forecasts; these findings underscore both the value and the difficulty of predictor selection in forecasting. When predictors are individually weak but numerous, multiple supervised-learning approaches that scale and screen factors before estimation can recover forecastable signal that single-pass selection discards (Hounyo and Li, 2026), and shrinkage methods are particularly effective in such data-rich environments (Dai et al., 2022).

特征选择是高维预测的核心问题。Guyon和Elisseeff (2003)提供了变量与特征选择

的经典论述，强调相关性准则的选择应紧扣学习任务。在惩罚方法中，lasso (Tibshirani, 1996)施加ℓ_1惩罚并将系数收缩至零，构成了稀疏线性特征选择基准。在金融领域，Welch和Goyal (2008)与Rapach等(2010)证明单个预测变量在样本外并不稳健，而跨预测变量组合信息能够改善股票溢价预测；这些发现既凸显了预测中变量选择的价值，也凸显了其难度。当预测变量单独微弱但数量众多时，在估计前对因子进行缩放与筛选的多重监督学习方法能够恢复单次筛选所丢弃的可预测信号 (Hounyo and Li, 2026)，而收缩方法在此类数据丰富环境中尤为有效 (Dai et al., 2022)。

A complementary response to predictor instability is forecast combination. Rather than betting on a single model, combining forecasts across model classes delivers robust gains (Wang et al., 2023; Qian et al., 2022; Greenaway-McGrevy, 2022), and imposing non-negativity and shrinkage on the combination weights further improves robustness in unstable environments (Roccazzella et al., 2022). Ensemble and blending architectures have likewise advanced financial forecasting, from decomposition-ensemble designs for commodity prices with multisource heterogeneous data (Zhao et al., 2026) to blending ensembles with interpretability diagnostics for tail risk (Ma and Zhang, 2026). At the same time, the benchmark side matters as much as the model side: simple, well-specified statistical benchmarks remain remarkably hard to beat even in the machine-learning era, and evaluation design choices such as window length can dominate model choice (Chassot and Audrino, 2026); forecast accuracy after extreme observations deteriorates in ways that favor conservative, anchored forecasts (Boug et al., 2026). We therefore treat the random walk not merely as a competitor but as an explicit *anchor column* inside the combination, so that the ensemble can collapse toward the benchmark whenever the estimated weights carry no signal.

对预测变量不稳定性的一种互补回应是预测组合。与其押注单一模型，跨模型类别组合预测能带来稳健增益 (Wang et al., 2023; Qian et al., 2022; Greenaway-McGrevy, 2022)，而对组合权重施加非负约束与收缩能在不稳定环境中进一步提升稳健性 (Roccazzella et al., 2022)。集成与混合架构同样推动了金融预测的发展，从基于多源异构数据的商品价格分解-集成设计 (Zhao et al., 2026)，到带可解释性诊断的尾部风险混合集成 (Ma and Zhang, 2026)。与此同时，基准一侧与模型一侧同等重要：设定恰当的简单统计基准即便在机器学习时代依然难以击败，窗口长度等评估设计选择甚至可能压倒模型选择 (Chassot and Audrino, 2026)；极端观测之后的预测精度会以有利于保守锚定预测的方式恶化 (Boug et al., 2026)。因此，本文不把随机游走仅视为竞争者，而是将其作为显式的*锚定列*纳入组合内部，使集成在估计权重不含信号时能够坍缩回基准。

The random-walk benchmark has a particularly long-standing role in financial forecasting. Meese and Rogoff (1983) famously showed that structural exchange-rate models cannot outperform a random walk out of sample, and this "Meese–Rogoff puzzle" has shaped the forecasting literature ever since; even sentiment-augmented models of exchange rates deliver improvements that are intermittent and regime-dependent (Ito and Takeda, 2022). In this paper we adopt the no-drift random walk as the classic financial benchmark, and we use the Diebold–Mariano (DM) test (Diebold and Mariano, 1995; Harvey et al., 1997) to establish whether any forecast improvement over the random walk is statistically significant, together with the Pesaran–Timmermann (PT) test (Pesaran and Timmermann, 1992, 2009) for directional predictability; because our claims span four horizons simultaneously, we report the DM statistic horizon by horizon in the spirit of multi-horizon forecast evaluation (Grant et al., 2026).

随机游走基准在金融预测中拥有特别悠久的地位。Meese和Rogoff (1983)的著名研究表明，结构性汇率模型无法在样本外击败随机游走，这一"Meese–Rogoff难题"自此塑造了整个预测文献；即便加入情绪指标的汇率模型，其改进也是间歇且依赖体制的 (Ito and Takeda, 2022)。本文采用无漂移随机游走作为经典金融基准，并使用Diebold–Mariano（DM）检验 (Diebold and Mariano, 1995; Harvey et al., 1997)来确立相对随机游走的预测改进是否统计显著，同时使用Pesaran–Timmermann（PT）检验 (Pesaran and Timmermann, 1992, 2009)检验方向可预测性；由于本文的结论同时横跨四个期限，我们遵循多期限预测评估的精神 (Grant et al., 2026)逐期限报告DM统计量。

2.6. Explainable AI and SHAP in Finance / 金融中的可解释人工智能与*SHAP*

The black-box nature of deep models has motivated a large XAI literature. SHAP (Lundberg and Lee, 2017) unifies several attribution methods by computing Shapley values (Štrumbelj and Kononenko, 2014) from cooperative game theory, providing additive, locally accurate, and consistent feature attributions. Lundberg et al. (2020) further extend SHAP to tree ensembles and show how local explanations can be aggregated into global understanding. In finance, SHAP has been used predominantly as a *post-hoc* explanation tool for trained models, although recent interpretable machine-learning studies increasingly couple explanation with multi-horizon forecasting tasks (Dong and Li, 2026; Ma and Zhang, 2026). A key methodological question—whether the information contained in SHAP rankings can be used *before* final model estimation to select features and improve out-of-sample prediction—has received far less attention. The closest precedents come from cryptocurrency and multi-horizon forecasting: El Youssefi et al. (2025) show that ex-ante feature selection can discard the vast majority of technical indicators without sacrificing walk-forward accuracy in cryptocurrency markets, and Dong and Li (2026) couple interpretable selection with multi-horizon risk forecasting. We extend this line of work in three directions: we use SHAP rankings as the ex-ante selector (SHAP-FS), benchmark it against both an all-feature and a LASSO-selected specification inside a nested walk-forward design, and—critically—embed the selected strategies within an anchored forecast-combination layer rather than treating selection as the final modelling step.

深度模型的黑箱性质催生了大量的XAI文献。SHAP (Lundberg and Lee, 2017)通过计算合作博弈论中的Shapley值 (Štrumbelj and Kononenko, 2014)统一了多种归因方法，提供加性、局部精确且一致的特征归因。Lundberg等(2020)进一步将SHAP扩展到树集成，并展示了如何将局部解释聚合为全局理解。在金融领域，SHAP主要被用作已训练模型的事后解释工具，近期可解释机器学习研究也开始将解释与多期限预测任务结合 (Dong and Li, 2026; Ma and Zhang, 2026)。一个关键的方法论问题——SHAP排序所蕴含的信息能否在最终模型估计之前被用于选择特征并改善样本外预测——受到的关注要少得多。最接近的先例来自加密货币与多期限预测：El Youssefi et al. (2025)证明事前特征选择可以在不牺牲前向验证精度的前提下剔除绝大部分技术指标，Dong and Li (2026)则将可解释选择与多期限风险预测相结合。本文在三个方向上扩展这一脉络：以SHAP排序作为事前选择器（SHAP-FS）；在嵌套前向设计内将其与全特征、LASSO筛选设定同时对标；并且——关键地——将所选策略嵌入锚定的预测组合层，而非把特征选择当作建模的最后一步。

2.7. *Research Gaps and Positioning* / 研究缺口与本文定位

In sum, the literature leaves four gaps that this paper addresses. First, no framework has tested whether feature-based forecasts of stablecoin issuance are statistically robust against the random walk in a properly nested design (Gap 1). Second, given the richness of on-chain, derivatives, macro, and sentiment data, predictor selection that preserves nonlinear signal while avoiding structural overfitting remains unresolved for issuance forecasting (Gap 2). Third, SHAP has rarely been validated as an *ex-ante* feature selector for forecasting rather than an *ex-post* report (Gap 3). Fourth, systematic evidence on how issuance drivers differ across horizons and across stablecoin types is lacking (Gap 4). Our paper fills these gaps by combining ex-ante SHAP-FS, LSTM/Transformer forecasting, and a horizon-by-stablecoin-type driver attribution framework.

综上，文献留下四个本文着手解决的缺口。第一，尚无框架在规范的嵌套设计中检验基于特征的稳定币发行量预测相对随机游走是否统计稳健（缺口1）。第二，鉴于链上、衍生品、宏观与情绪数据的丰富性，在保留非线性信号的同时避免结构性过拟合的预测变量选择在发行预测中仍未解决（缺口2）。第三，SHAP很少被验证为预测的事前特征选择器，而更多地作为事后报告（缺口3）。第四，关于发行驱动因素在不同

期限及不同稳定币类型之间如何差异的系统证据仍然缺乏（缺口4）。本文通过结合事前SHAP-FS、LSTM/Transformer预测以及期限×稳定币类型驱动归因框架来填补这些缺口。

3. Research Framework / 研究框架

Our research framework follows a three-stage empirical design, in the spirit of walk-forward ensemble forecasting for cryptocurrencies (Oyewola et al., 2022), as illustrated in Figure 1.

本文采用三阶段实证设计的研究框架，与加密货币前向集成预测实践 (Oyewola et al., 2022) 一脉相承，如图 1 所示。

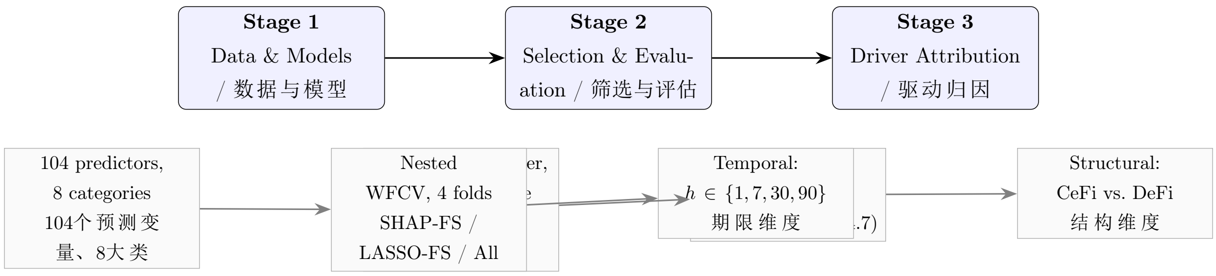


图 1: Three-Stage Research Framework / 三阶段研究框架

Stage 1: Data Preparation and Model Construction / *阶段1*：*数据准备与模型构建*. We collect and preprocess daily data for three major stablecoins (USDT, USDC, and DAI) and construct a unified panel of **104** initial candidate predictors spanning eight categories: macro-external (M, **12**), Fed liquidity plumbing (F, **8**), crypto-global (G, 19), BTC derivatives leverage (L, 28), exchange flows (E, 11), depeg and arbitrage (P, 3), on-chain structure (S, 19), and autoregressive momentum (A, 4). The F block is drawn from the Federal Reserve's FRED database—the Secured Overnight Financing Rate (SOFR), overnight reverse repurchase agreement (ON RRP) volume, the Treasury General Account (TGA) balance, and total Fed balance-sheet assets, from which we construct a net-liquidity measure (Fed assets minus TGA minus ON RRP)—with each series entering in both level and first-difference form and aligned to its publication lag (daily series lagged one day, the weekly balance-sheet series lagged eight days) to prevent look-ahead. We build the core forecasting models—LSTM as the primary forecaster and Transformer as the deep-sequence benchmark—together with random forest, XGBoost, a no-drift random walk, and a **random walk with drift** as benchmarks.

我们收集并预处理三种主要稳定币（USDT、USDC与DAI）的日度数据，构建涵盖八个类别的**104**个初始候选预测变量的统一面板：宏观外部（M类，**12**个）、美联储流动性管道（F类，**8**个）、加密全局（G类，19个）、BTC衍生品杠杆（L类，28个）、交易所资金流（E类，11个）、脱锚套利（P类，3个）、链上结构（S类，19个）与自回归动量（A类，4个）。F类取自美联储FRED数据库——担保隔夜融资利率（SOFR）、隔夜逆回购（ON RRP）规模、财政部一般账户（TGA）余额与美联储资产负债表总规模，并由此构建净流动性指标（美联储资产减去TGA再减去ON RRP）——每个序列同时以水平值与一阶差分形式进入，且按发布滞后对齐（日度序列滞后1天、周度资产负债表序列滞后8天）以杜绝前瞻。我们构建核心预测模型——以LSTM为主预测器、Transformer为深度序列基准——并辅以随机森林、XGBoost、**无漂移随机游走与带漂移随机游走**作为基准。

Stage 2: Feature Selection and Forecast Evaluation / *阶段2*：*特征选择与预测评估*. We compare three feature-selection strategies—All-Features, LASSO-FS (linear baseline), and SHAP-FS (our proposed method)—under a nested walk-forward cross-validation with four folds and approximately 12-month blind test periods (6 months in the final fold). For each strategy, the individual forecasts are then passed through a forecast-combination layer that (a) estimates non-negative stacking weights by horizon from inner-loop validation predictions only, with the random walk included as an explicit anchor, and (b) applies a conservative global shrinkage toward the no-change forecast (Section 4.7). We evaluate all models using RMSE, MAE, the Pesaran–Timmermann directional test, the out-of-sample R^2 against the random walk, and the Diebold–Mariano test against the random walk, covering 3 stablecoins $\times$ 4 horizons = 12 prediction tasks.

我们在四折、12个月盲测期的嵌套前向交叉验证下比较三种特征选择策略——全特征、LASSO-FS（线性基准）与SHAP-FS（本文方法）。随后，每种策略的单个预测会经过一个预测组合层：（a）仅利用内循环验证集预测按期限估计非负堆叠权重，并将随机游走作为显式锚纳入；（b）施加朝向无变化预测的保守全局收缩（第 4.7 节）。我们使用RMSE、MAE、Pesaran–Timmermann方向检验、相对随机游走的样本外R^2以及相对随机游走的Diebold–Mariano检验评估所有模型，覆盖3种稳定币×4个期限= 12个预测任务。

Stage 3: Driver Heterogeneity Analysis / *阶段3*：*驱动因素异质性分析*. Using the out-of-sample SHAP values extracted in Stage 2, we map the heterogeneity of issuance drivers along two dimensions: the temporal dimension (ultra-short-term 1-day vs. short-term 1-week vs. medium-term 1-month vs. long-term 3-month) and the structural dimension (centralized USDT/USDC vs. decentralized DAI).

利用阶段2提取的样本外SHAP值，我们沿两个维度刻画发行驱动因素的异质性：时间维度（超短期1天 vs 短期1周 vs 中期1月 vs 长期3月）与结构维度（中心化USDT/USDC vs 去中心化DAI）。

4. Data and Methodology / 数据与方法

4.1. Sample Design / 样本设计

We use daily data from January 1, 2020, to June 30, 2026 (about 2,370 observations per series) for three major stablecoins: USDT (centralized, fiat-collateralized), USDC (centralized, regulation-oriented), and DAI (decentralized, crypto-overcollateralized). The prediction target is the log change rate of stablecoin issuance over a horizon h:

我们使用2020年1月1日至2026年6月30日的日度数据（每序列约2370个观测）研究三种主要稳定币：USDT（中心化、法币抵押型）、USDC（中心化、合规导向型）与DAI（去中心化、加密资产超额抵押型）。预测目标为未来h期稳定币发行量的对数变化率：

$$y_t^{(i,h)} = \ln\big(S_{t+h}^{(i)}\big) - \ln\big(S_t^{(i)}\big) = \sum_{k=1}^{h} \Delta \ln S_{t+k}^{(i)}, \tag{1}$$

where $S_t^{(i)}$ is the circulating supply (market capitalization) of stablecoin $i \in \{$USDT, USDC, DAI$\}$ at day t, and $h \in \{1, 7, 30, 90\}$ denotes the forecast horizon in days (1-day ultra-short-term, 1-week short-term, 1-month medium-term, and 3-month long-term). This transformation yields an approximately stationary target series, avoids the spurious-regression problem associated with non-stationary level series, and makes RMSE/MAE comparable across stablecoins.

其中$S_t^{(i)}$为稳定币$i \in \{$USDT、USDC、DAI$\}$在第t天的流通供应量（市值），$h \in \{1, 7, 30, 90\}$表示以天计的预测期限（1天超短期、1周短期、1月中期、3月长期）。该变换得到近似平稳的目标序列，避免非平稳水平序列带来的伪回归问题，并使RMSE/MAE在不同稳定币之间可比。

Because crypto markets move far faster than foreign-exchange markets, a daily sampling frequency allows us to form a complete spectrum from high-frequency trading signals (1 day) to medium- and long-term fundamentals (1 week to 3 months), in line with the multi-horizon forecasting design of Dong and Li (2026).

由于加密市场变化远快于外汇市场，日度采样频率使我们得以构成从高频交易信号（1天）到中长期基本面（1周至3月）的完整期限谱系，与Dong and Li (2026)的多期限预测设计一致。

4.2. Predictor Pool / 预测变量池

Starting from the eight-category panel described in Stage 1, the initial feature pool contains **104** candidate predictors (100 external candidate variables from the M/F/G/L/E/P/S categories plus 4 autoregressive terms from the A category), detailed in the appendix (附录 A). All variables follow a unified transformation principle—*flow variables are matched to the flow target, ratio variables retain levels, and stock variables enter as change rates*—to align their stationarity with the target variable in Eq. (1):

如阶段1所述，从八类别面板出发，初始特征池包含**104**个候选预测变量（M/F/G/L/E/P/S类别的**100**个外部候选变量加A类别的4个自回归项），详见附录（附录 A）。所有变量遵循统一的变换原则——*流量对流量，比率留原值，存量取变化率*——以使其平稳性与式(1)的目标变量一致：

$$X_{j,t} = \begin{cases} \Delta \ln X_{j,t}^{raw}, & \text{stock-type variables} \\ & \text{(price, market cap, OI, TVL, addresses, balances),} \\ X_{j,t}^{raw}, & \text{ratio/rate/index variables} \\ & \text{(funding rate, basis, volatility, premium, ratios),} \\ z_t\big(\ln(1 + X_{j,t}^{raw})\big), & \text{spiky flow variables} \\ & \text{(liquidations, large transfers),} \\ \big\{X_{j,t}^{raw},\ \Delta X_{j,t}^{raw}\big\}, & \text{macro interest rates} \\ & \text{and policy-risk indices,} \end{cases} \tag{2}$$

where $\Delta X_{j,t} = X_{j,t} - X_{j,t-1}$ denotes the first difference, and $z_t(\cdot)$ denotes the rolling 30-day z-score. A handful of macro rate/policy series and the four Fed liquidity-plumbing series are retained in both level and first-difference form, so that the standardized modeling panel contains 104 columns. Neural-network inputs are further normalized to $[0, 1]$ by a min–max transformation that is fitted only on the inner-loop training window, $\mathcal{T}_{tr}$:

其中$\Delta X_{j,t} = X_{j,t} - X_{j,t-1}$表示一阶差分，$z_t(\cdot)$表示30天滚动z-score。少量宏观利率/政策序列与四个美联储流动性管道序列同时保留水平值与一阶差分形式，因此标准化建模面板共104列。神经网络输入进一步通过仅在训练窗口$\mathcal{T}_{tr}$上拟合的min-max变换归一化到$[0, 1]$：

$$\widetilde{X}_{j,t} = \frac{X_{j,t} - \min_{\tau \in \mathcal{T}_{tr}} X_{j,\tau}}{\max_{\tau \in \mathcal{T}_{tr}} X_{j,\tau} - \min_{\tau \in \mathcal{T}_{tr}} X_{j,\tau}}. \tag{3}$$

Fitting the normalizer only within the training window is a crucial step for avoiding look-ahead bias. Table 2 summarizes the eight categories and their economic rationale;

variable-level definitions and transformation formulas are provided in Appendix 附录 A.

仅训练窗口内拟合标准化器是避免前瞻偏差的关键一步。表 2 概括了八个类别及其经济逻辑；变量级定义与变换公式见附录 附录 A。

This large initial pool deliberately avoids excluding predictors ex ante; the SHAP-FS procedure automatically selects the informative subset that optimizes out-of-sample prediction for each task.

这个庞大的初始池有意识地避免事前排除预测变量；SHAP-FS流程将为每个任务自动选择优化样本外预测的信息子集。

4.3. Core Models / 核心模型

We build two deep sequence models and compare them against a set of penalized-linear, tree-based, and random-walk benchmarks, which later enter the forecast-combination layer of Section 4.7.

我们构建两种深度序列模型，并与一组惩罚线性、树模型及随机游走基准进行比较；这些模型随后都会进入第 4.7 节的预测组合层。

4.3.1. LSTM / LSTM

The LSTM network (Hochreiter and Schmidhuber, 1997) is designed to learn long-term dependencies through gated memory cells. At each time step t, given the input x_t and the previous hidden state h_{t-1}, the input gate i_t, forget gate f_t, output gate o_t, cell candidate $\tilde{c}_t$, cell state c_t, and hidden state h_t are updated as

LSTM网络 (Hochreiter and Schmidhuber, 1997)通过带门的记忆单元学习长期依赖。在每个时间步t，给定输入x_t与前一隐藏状态h_{t-1}，输入门i_t、遗忘门f_t、输出门o_t、候选单元$\tilde{c}_t$、单元状态c_t与隐藏状态h_t按如下方式更新：

$$i_t = \sigma\big(W_i x_t + U_i h_{t-1} + b_i\big), \tag{4a}$$

$$f_t = \sigma\big(W_f x_t + U_f h_{t-1} + b_f\big), \tag{4b}$$

$$o_t = \sigma\big(W_o x_t + U_o h_{t-1} + b_o\big), \tag{4c}$$

$$\tilde{c}_t = \tanh\big(W_c x_t + U_c h_{t-1} + b_c\big), \tag{4d}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{4e}$$

$$h_t = o_t \odot \tanh(c_t), \tag{4f}$$

where $\sigma(\cdot)$ is the logistic sigmoid, $\odot$ denotes element-wise multiplication, and W, U, b are trainable parameters. A regression head maps the final hidden state to the horizon-h issuance change rate. The input at each time step is the vector of predictors, so the

表 2: Summary of the Initial Predictor Pool by Category | 初始预测变量池分类汇总

Category	Description	# Vars	Economic rationale
M	Macro-external market	**12**	Reserve-yield source, dollar-liquidity conditions, policy/geopolitical risk, global risk appetite
F	Fed liquidity plumbing	**8**	SOFR, ON RRP volume, TGA balance, Fed balance sheet (net liquidity); the QT/QE–reserves channel into stablecoin demand
G	Crypto-global indicators	19	Market scale, leverage stock, valuation regime, sentiment, USDT OTC premium
L	BTC derivatives leverage	28	Funding rates, open interest, liquidations, volumes, options; the leverage–stablecoin channel
E	Exchange flows	11	Inflows/outflows, balances, deposit/withdrawal breadth; mint/redeem at exchanges
P	Depeg and arbitrage	3	Secondary-market deviation from peg drives primary-market mint/redeem
S	On-chain structure	19	Addresses, transfers, holder behavior, concentration; on-chain demand microstructure
A	Autoregressive and momentum	4	Lagged issuance changes, rolling mean and volatility; forecasting baseline
Total			**104**

Notes: The 104 candidate predictors are composed of 100 external variables plus 4 autoregressive terms; three macro series and four Fed liquidity-plumbing series are retained in both level and first-difference form.

类别	描述	变量数	经济逻辑
M	宏观与外部市场	**12**	储备收益来源、美元流动性状况、政策/地缘政治风险、全球风险偏好
F	美联储流动性管道	**8**	SOFR、ON RRP规模、TGA余额、美联储资产负债表（净流动性）；量化宽松/紧缩向稳定币需求的储备传导渠道
G	加密全局指标	19	市场规模、杠杆存量、估值体制、情绪、USDT场外溢价
L	BTC衍生品杠杆	28	资金费率、持仓量、清算量、成交量、期权；杠杆—稳定币渠道
E	交易所资金流	11	流入/流出、余额、存取款广度；交易所铸造/赎回
P	脱锚套利	3	二级市场偏离锚定驱动一级市场铸造/赎回
S	链上结构	19	地址数、转账、持有者行为、集中度；链上需求微观结构
A	自回归动量	4	滞后发行量变化、滚动均值与波动率；预测基线
合计			**104**

注：104个候选预测变量由100个外部变量加4个自回归项构成；三个宏观序列与四个美联储流动性管道序列同时保留原值与一阶差分。

model sees a three-dimensional tensor of shape $N \times L \times F$, where L is the lookback window (30/60/90 days, tuned in the inner loop) and F is the number of features.

其中$\sigma(\cdot)$为逻辑斯蒂sigmoid函数，$\odot$表示逐元素相乘，W、U、b为可训练参数。回归头将最终隐藏状态映射为h期发行量变化率。每个时间步的输入为预测变量向量，因此模型接受形状为$N \times L \times F$的三维张量，其中L为回溯窗口（30/60/90天，内循环调优）、F为特征数。

4.3.2. Transformer Encoder / Transformer编码器

The Transformer encoder (Vaswani et al., 2017) replaces recurrence with multi-head self-attention. Let Q, K, V denote the query, key, and value matrices; scaled dot-product attention is

Transformer编码器 (Vaswani et al., 2017)以多头自注意力替代递归。令Q、K、V分别表示查询、键与值矩阵，缩放点积注意力为

$$\mathrm{Attention}(Q, K, V) \;=\; \mathrm{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V, \tag{5}$$

where d_k is the key dimension. In our implementation, the raw predictor sequence is linearly projected, augmented with learned positional encodings, passed through a stack of Transformer encoder layers, and pooled across time before the regression head. This architecture lets the model attend to global dependencies at any time lag, complementing the LSTM's sequential inductive bias.

其中d_k为键的维度。在我们的实现中，原始预测变量序列经线性投影、叠加可学习位置编码后，通过多层Transformer编码器，并在回归头前跨时间池化。该架构使模型能够在任意时间滞后上关注全局依赖，与LSTM的序列归纳偏置互补。

4.3.3. Benchmark Models / 基准模型

We compare the deep models against **five** benchmarks and statistical anchors: (i) **Random Forest**, a tree-ensemble method handling nonlinearities and feature interactions; (ii) **XGBoost**, a gradient-boosted tree method widely used in financial forecasting (Jaquart et al., 2021); (iii) **ridge regression**, a penalized linear benchmark on the same feature set; (iv) the **random walk without drift**, the classic financial benchmark (Meese and Rogoff, 1983), which forecasts the horizon-h change rate as zero (i.e., supply stays at its current level); and (v) two drift-type anchors—the **random walk with drift**, which forecasts the horizon-h change rate as the mean of the change rates observed in the training window, and an **exponentially weighted moving-average (EMA) drift** forecast that weights recent changes more heavily. To stabilize the deep learners, the LSTM is averaged over three random seeds and the Transformer over two; seed averaging alone materially reduces the variance of deep forecasts in our experiment.

我们将深度模型与**五个**基准及统计锚比较：（i）**随机森林**，一种处理非线性和特征交互的树集成方法；（ii）**XGBoost**，一种在金融预测中广泛使用的梯度提升树方法 (Jaquart et al., 2021)；（iii）**岭回归**，基于同一特征集的惩罚线性基准；（iv）**无漂移随机游走**，经典金融基准 (Meese and Rogoff, 1983)，其将h期变化率预测为零（即供应量保持当前水平）；（v）两个漂移型锚——**带漂移随机游走**，将h期变化率预测为训练窗口内变化率均值，以及**指数加权移动平均（EMA）漂移**预测，对近期变化赋予更高权重。为稳定深度学习器，LSTM对三个随机种子取平均、Transformer对两个种子取平均；在我们的实验中，仅种子平均一项就显著降低了深度预测的方差。

4.4. Nested Walk-Forward Cross-Validation / 嵌套前向交叉验证

All feature transformations, hyperparameter tuning, feature selection, and evaluation are embedded in a nested walk-forward cross-validation (WFCV) design to eliminate look-ahead bias, as illustrated in Figure 2. The outer loop comprises four folds with an expanding training window and a completely untouched blind test set per fold (12 months for the first three folds and 6 months for the last):

所有特征变换、超参数调优、特征选择与评估均嵌入嵌套前向交叉验证（WFCV）设计以消除前瞻偏差，如图 2 所示。外循环包含四折，每折使用扩展训练窗口并保留一个完全未触碰的盲测集（前三折各12个月，最后一折6个月）：

表 3: Outer-Loop Folds of the Nested Walk-Forward Cross-Validation | 嵌套前向交叉验证的外循环折叠

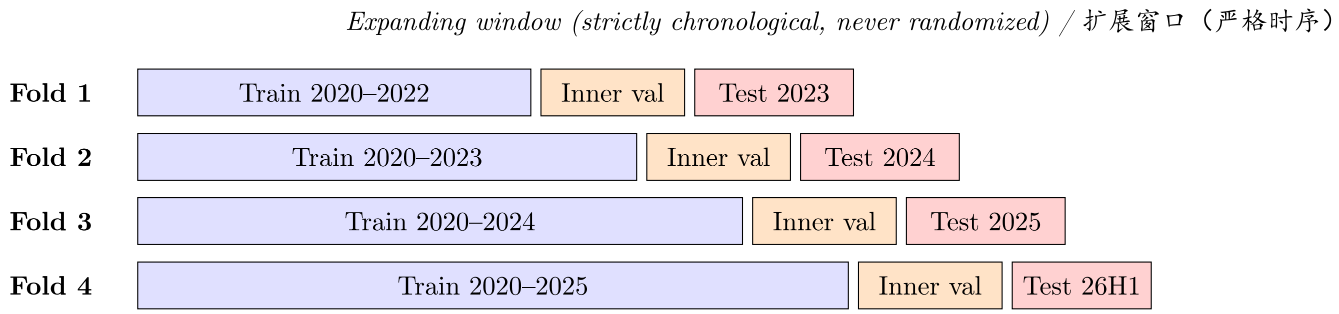
Fold	**Training window**	**Blind test**	**Notable events**
Fold 1	2020-01-01 – 2022-12-31	2023 (12 months)	Silicon Valley Bank (USDC de-peg)
Fold 2	2020-01-01 – 2023-12-31	2024 (12 months)	BTC spot ETF approval
Fold 3	2020-01-01 – 2024-12-31	2025 (12 months)	Interest-rate cycle
Fold 4	2020-01-01 – 2025-12-31	2026H1 (6 months)	–

Notes: The inner loop uses the last 365 days of each fold's training window as the inner validation set. All partitions are strictly chronological and never randomized.

The validation design follows three key rules: (i) **no look-ahead**—all feature transformations, parameter tuning, and feature selection use only information available before the test period; (ii) **expanding window**—the training window expands fold by fold to

折	训练窗口	盲测期	关键事件
第1折	2020-01-01 – 2022-12-31	2023（12个月）	硅谷银行事件（USDC脱锚）
第2折	2020-01-01 – 2023-12-31	2024（12个月）	BTC现货ETF获批
第3折	2020-01-01 – 2024-12-31	2025（12个月）	利率周期
第4折	2020-01-01 – 2025-12-31	2026上半年（6个月）	–

注：内循环以每折训练窗口尾部365天作为内验证集；所有划分严格保持时间顺序，不随机化。



Inner loop (blue+orange): hyperparameter tuning, SHAP ranking & subset selection, and ensemble-weight estimation use only pre-test data; the red blind test is touched exactly once. An h-day embargo separates inner validation from each test block.
内循环（蓝+橙）：超参调优、SHAP排序与子集选择、集成权重估计仅使用测试期之前的数据；红色盲测集只触碰一次；内验证与测试块之间设有h天禁运。

图 2: Nested Walk-Forward Cross-Validation Design / 嵌套前向交叉验证设计

incorporate new information; and (iii) **blind testing**—the outer-loop test set is never accessed during inner-loop tuning and selection. The key principle is: *model choice is tuned inside the inner loop; forecasting claims are evaluated outside in the outer loop.*

验证设计遵循三个关键规则：（i）**无前瞻**——所有特征变换、参数调优与特征选择仅使用测试期之前可得的信息；（ii）**扩展窗口**——训练窗口逐折扩展以吸收新信息；（iii）**盲测**——外循环测试集在内循环调优与选择过程中绝不访问。关键原则是：*模型选择在内循环完成，预测声明在外循环检验。*

4.5. Evaluation Metrics / 评估指标

We evaluate prediction performance along four dimensions, following standard practice in the machine-learning forecasting literature (Ma and Zhang, 2026; Dong and Li, 2026): the root mean squared error (RMSE) and the mean absolute error (MAE),

我们遵循机器学习预测文献的标准实践 (Ma and Zhang, 2026; Dong and Li, 2026)，从四个维度评估预测性能：均方根误差（RMSE）与平均绝对误差（MAE），

$$\text{RMSE} \;=\; \sqrt{\frac{1}{N}\sum_{i=1}^{N}\big(y_i-\hat{y}_i\big)^2}, \qquad \text{MAE} \;=\; \frac{1}{N}\sum_{i=1}^{N}\big|y_i-\hat{y}_i\big|; \tag{6}$$

the Pesaran–Timmermann (PT) test for directional predictability (Pesaran and Timmermann, 1992, 2009),

方向可预测性的Pesaran–Timmermann（PT）检验 (Pesaran and Timmermann, 1992, 2009)，

$$S_{PT} \;=\; \frac{\hat{P}-\hat{P}^*}{\sqrt{\hat{P}^*\big(1-\hat{P}^*\big)/N}}, \qquad \hat{P}^* \;=\; \hat{p}_y\,\hat{p}_{\hat{y}}+\big(1-\hat{p}_y\big)\big(1-\hat{p}_{\hat{y}}\big), \tag{7}$$

where $\hat{P}$ is the observed directional hit rate, $\hat{p}_y$ and $\hat{p}_{\hat{y}}$ are the sample frequencies of positive outcomes in the realized and predicted series, and $\hat{P}^*$ is the expected hit rate under independence; and the Diebold–Mariano (DM) test (Diebold and Mariano, 1995; Harvey et al., 1997) for the statistical significance of the difference between two forecasts,

其中$\hat{P}$为观测到的方向命中率，$\hat{p}_y$与$\hat{p}_{\hat{y}}$为实际与预测序列中正结果样本频率，$\hat{P}^*$为独立假设下的期望命中率；以及用于比较两个预测差异显著性的Diebold–Mariano（DM）检验 (Diebold and Mariano, 1995; Harvey et al., 1997)，

$$S_{DM} \;=\; \frac{\bar{d}}{\sqrt{\widehat{V}(\bar{d})}}, \qquad \widehat{V}(\bar{d}) \;=\; \frac{1}{N}\Big[\hat{\gamma}_0+2\sum_{k=1}^{h-1}\omega_k\,\hat{\gamma}_k\Big], \qquad \omega_k \;=\; 1-\frac{k}{h}, \tag{8}$$

where $d_t = e_{1,t}^2 - e_{2,t}^2$ is the loss-differential series built on squared errors, $\bar{d}$ is its mean, $\hat{\gamma}_k$ is the k-th sample autocovariance, and ω_k are Bartlett weights with lag truncated to $\min(h-1, N^{1/4})$ to avoid non-positive HAC variance estimates in small samples. We also apply the Harvey–Leybourne–Newbold (HLN) finite-sample correction. A negative S_{DM} indicates that model 1 is more accurate; we report both the random-walk comparisons (RW=0 and RW-drift) for every model and a direct SHAP-FS vs. LASSO-FS comparison for the LSTM.

其中$d_t = e_{1,t}^2 - e_{2,t}^2$为基于平方误差的损失差分序列，$\bar{d}$为其均值，$\hat{\gamma}_k$为第$k$阶样本自协方差，$\omega_k$为Bartlett权重，滞后取$\min(h-1, N^{1/4})$自动截断，避免小样本下HAC方差估计非正，并加入Harvey–Leybourne–Newbold（HLN）小样本修正。$S_{DM} < 0$表示模型1更精确；我们对每个模型报告相对随机游走的比较（RW=0与RW-drift），并对LSTM报告SHAP-FS vs LASSO-FS的直接DM比较。

4.6. SHAP-FS: Ex-Ante Feature Selection / SHAP-FS：事前特征选择

The methodological core of this paper is repositioning SHAP from a post-hoc explanation tool to an ex-ante feature selector (SHAP-FS). Unlike linear selection rules such as the lasso (Tibshirani, 1996), SHAP-FS derives feature importance from a trained nonlinear sequence model, thereby preserving the complex nonlinear signals and interaction effects that a linear pre-filter would discard.

本文的方法论核心是将SHAP从事后解释工具重新定位为事前特征选择器（SHAP-FS）。与lasso (Tibshirani, 1996)等线性选择规则不同，SHAP-FS从训练好的非线性序列模型中推导特征重要性，从而保留线性预筛会丢弃的复杂非线性信号与交互效应。

*4.6.1. SHAP Values / SHAP*值

For a model f and a set of features $\mathcal{F}$ with M predictors, the SHAP value $\phi_{j,i}$ of feature j for observation i is the Shapley-value contribution (Štrumbelj and Kononenko, 2014; Lundberg and Lee, 2017):

对模型f与含M个预测变量的特征集合$\mathcal{F}$，特征j对观测i的SHAP值$\phi_{j,i}$是Shapley值贡献（Štrumbelj and Kononenko, 2014; Lundberg and Lee, 2017)：

$$\phi_{j,i} = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!\,\big(M - |S| - 1\big)!}{M!} \Big[f_x\big(S \cup \{j\}\big) - f_x(S) \Big], \tag{9}$$

where $f_x(S)$ is the model prediction conditional on the feature subset S. The additive attribution property $\sum_j \phi_{j,i} = f(x_i) - \mathbb{E}[f]$ makes SHAP values locally accurate and consistent. We compute SHAP values for the LSTM using a gradient-based expected-gradients estimator; for tree models we use the TreeSHAP algorithm (Lundberg et al., 2020). The out-of-sample attributions reported in the heterogeneity analysis average the TreeSHAP values of the SHAP-FS strategy's random-forest and XGBoost learners on each task's blind test window.

其中$f_x(S)$为在特征子集S条件下模型的预测。加性归因性质$\sum_j \phi_{j,i} = f(x_i) - \mathbb{E}[f]$使SHAP值局部精确且一致。对LSTM，我们使用基于梯度的期望梯度估计器计算SHAP值；对树模型，我们使用TreeSHAP算法 (Lundberg et al., 2020)。异质性分析报告的样本外归因，为每个任务盲测窗口上SHAP-FS策略随机森林与XGBoost学习器TreeSHAP值的平均。

The task-level global importance of each predictor is obtained by averaging absolute SHAP values across all outer-loop test samples,

各预测变量的任务级全局重要性通过对所有外循环测试样本的SHAP绝对值取平均得到，

$$I_j = \frac{1}{N}\sum_{i=1}^{N}\left|\phi_{j,i}\right|. \tag{10}$$

4.6.2. The Five-Phase SHAP-FS Procedure / SHAP-FS五阶段流程

The SHAP-FS procedure strictly separates direction-finding, subset-size selection, final retuning, blind evaluation, and SHAP extraction, and nests them within the cross-validation loops:

SHAP-FS流程将方向寻找、子集规模选择、最终调优、盲评估与SHAP提取五件事严格分离，并全部嵌套于交叉验证循环之中：

1. **Phase 1: Initial tuning and SHAP ranking / 阶段1：初始调优与SHAP排序**. Using all M features, we train an LSTM on the inner-loop training set with a grid search over hyperparameters (lookback window, hidden size, learning rate) and early stopping on the inner validation set, obtaining initial parameters Θ_{init}. We then compute global SHAP importance I_j via Eq. (10) and rank all predictors in descending order, producing the ranked list $\mathcal{R} = [f_{(1)}, f_{(2)}, \ldots, f_{(M)}]$.
   阶段1：初始调优与SHAP排序。使用全部M个特征，在内循环训练集上以网格搜索超参数（回溯窗口、隐层大小、学习率）并在内验证集上早停训练LSTM，得到初始参数Θ_{init}。随后通过式(10)计算全局SHAP重要性I_j，并按降序对所有预测变量排序，得到排序列表$\mathcal{R} = [f_{(1)}, f_{(2)}, \ldots, f_{(M)}]$。
2. **Phase 2: SHAP-FS forward screening / 阶段2：SHAP-FS前向筛选**. We sequentially test feature subsets of increasing size, $\mathcal{S}_k = \{f_{(1)}, \ldots, f_{(k)}\}$ for $k = 1, \ldots, M$, retraining the model with Θ_{init} and early stopping for each subset, and select the subset achieving the lowest inner-validation RMSE:

阶段2：SHAP-FS前向筛选。我们依次测试规模递增的特征子集$\mathcal{S}_k = \{f_{(1)}, \ldots, f_{(k)}\}$（$k = 1, \ldots, M$），对每个子集使用$\Theta_{init}$并早停重新训练，选择内验证RMSE最低的子集：

$$k^* = \arg \min_{k \in \{1,2,\ldots,M\}} \mathrm{RMSE}_{val}\big(\mathcal{S}_k\big). \tag{11}$$

3. **Phase 3: Final retuning / 阶段3：最终重调优**. With the optimal subset fixed, we perform a small-scale grid search to re-tune the model to the reduced feature space, obtaining final parameters Θ_{final}.
   阶段3：最终重调优。固定最优子集后，我们进行小范围网格搜索，将模型重新调优至降维后的特征空间，得到最终参数Θ_{final}。
4. **Phase 4: Outer-loop blind testing / 阶段4：外循环盲测**. We merge the inner-loop training and validation sets as the outer-loop training set. To prevent information leakage from the horizon-h overlap between Inner_Val and the blind test, the last h days of Inner_Val are excluded by an embargo. We then re-carve a validation tail from the merged outer-loop training set and re-apply early stopping, retrain the final model with Θ_{final} on the selected subset, and evaluate it exactly once on the untouched blind test set.
   阶段4：外循环盲测。我们将内循环训练集与验证集合并作为外循环训练集，使用Θ_{final}在选定子集上重新训练最终模型，并在未触碰的盲测集上仅评估一次。
5. **Phase 5: Local SHAP extraction / 阶段5：局部SHAP提取**. We extract local SHAP values on the blind test set; features not selected in the optimal subset are filled with zero to retain the full M-column SHAP matrix for the downstream heterogeneity analysis.
   阶段5：局部SHAP提取。我们在盲测集上提取局部SHAP值；对最优子集中未选中的特征补零列，以保留完整M列SHAP矩阵供后续异质性分析使用。

The nesting is indispensable: if SHAP-FS were run “train → rank → select → retrain → evaluate” on the same data, the SHAP ranking would already have “seen” the test set, constituting data leakage. The nested design ensures that outer-loop evaluations are unbiased, genuine out-of-sample performances.

嵌套结构必不可少：如果SHAP-FS在同一数据上“训练→排序→选特征→再训练→评估”，SHAP排序便已“看过”测试集，构成数据泄露。嵌套设计保证外循环评估是无偏的真实样本外表现。

4.6.3. SHAP-FS versus LASSO-FS / SHAP-FS与LASSO-FS的对比

Table 4 contrasts SHAP-FS with the linear LASSO-FS baseline along four dimensions.

表 4 从四个维度对比SHAP-FS与线性LASSO-FS基准。

表 4: SHAP-FS versus LASSO-FS | SHAP-FS与LASSO-FS的对比

Dimension	**LASSO-FS**	**SHAP-FS**
Selection rule	Linear ℓ_1 penalty shrinks coefficients to zero	SHAP values from a trained nonlinear sequence model
Nonlinear signals	May discard complex nonlinear predictors	Preserved (derived from a nonlinear model, capturing interactions)
Coupling with the forecaster	Disconnected linear pre-filter	Tightly coupled in-model selection
Performance across horizons	Limited or even adverse improvements	Improvements that widen monotonically with the horizon

维度	**LASSO特征选择**	**SHAP特征选择**
选择规则	线性ℓ_1惩罚将系数收缩至零	来自训练好的非线性序列模型的SHAP值
非线性信号	可能丢弃复杂非线性预测变量	得以保留（源自非线性模型，可捕捉交互效应）
与预测模型的耦合	脱节的线性预筛选	与模型紧耦合的选择
跨期限表现	改进有限甚至适得其反	随期限拉长而单调扩大的改进

4.7. From Feature Selection to Forecast Combination: The SHAP-FS Ensemble / 从特征选择到预测组合：SHAP-FS集成

As our results will show, no individual learner beats the no-drift random walk on its own. We therefore add a final combination layer that turns the SHAP-FS-selected models into a single anchored forecast, following the forecast-combination literature on constrained optimization and shrinkage (Roccazzella et al., 2022; Qian et al., 2022; Dai et al., 2022). The design rests on five pillars: a diversified base-learner zoo; horizon-pooled non-negative weight estimation conducted strictly inside the inner loop; equal-weight averaging of the three feature-selection strategies to diversify selector-specific estimation noise; validation walk-forward selection of the shrinkage coefficient and clipping bound; and evidence-based gating that collapses a task to the random walk whenever its inner-validation evidence is too weak.

正如后文结果所示，没有任何单个学习器能独自击败无漂移随机游走。因此，我们增设最后一层组合环节，将SHAP-FS选出的模型转化为单一锚定预测，遵循约束优化与收缩的预测组合文献 (Roccazzella et al., 2022; Qian et al., 2022; Dai et al., 2022)。该设计建立在五个支柱上：多元化的基学习器集合；严格在内循环内完成的按期限池化非负权重估计；对三种特征选择策略取等权平均以分散选择器特有的估计噪声；基于验证集前向走查选择收缩系数与截断边界；以及证据门控——当某任务的内验证证据过弱时将其坍缩回随机游走。

Base learners and anchors / 基学习器与锚. For each task (stablecoin i, horizon h, outer fold f) and each feature-selection strategy, the candidate pool contains: (i) the **LSTM**, averaged over three random seeds to stabilize the deep learner; (ii) the **Transformer encoder**, averaged over two seeds; (iii) **random forest**; (iv) **XGBoost** (plus its sign-classification variant, whose class probabilities enter as an additional directional forecast); (v) a **ridge regression** on the selected features; and three statistical anchors: the **no-drift random walk** (zero change), the **random walk with drift** (training-window mean change), and an **exponentially weighted moving-average drift** forecast. Including the anchors as candidate columns is essential: whenever the data-rich learners carry no signal, the combination can put all weight on the benchmark instead of being forced to extrapolate.

对每个任务（稳定币i、期限h、外折f）与每种特征选择策略，候选池包含：（i）**LSTM**，对三个随机种子取平均以稳定深度学习器；（ii）**Transformer编码器**，对两个种子取平均；（iii）**随机森林**；（iv）**XGBoost**（及其符号分类变体，类别概率作为额外的方向预测进入组合）；（v）基于所选特征的**岭回归**；以及三个统计锚：**无漂移随机游走**（零变化）、**带漂移随机游走**（训练窗口均值变化）与**指数加权移动平均漂移**预测。将锚作为候选列纳入至关重要：当数据驱动的学习器不含信号时，组合可以把全部权重交给基准，而非被迫外推。

Horizon-pooled non-negative stacking / 按期限池化的非负堆叠. Combination weights are estimated exclusively from inner-loop validation forecasts, so the outer-loop blind test remains untouched. Because the scale of issuance changes differs across stablecoins and folds, we standardize each task's inner-validation predictions and targets by the task-level target standard deviation σ_k before pooling. For each horizon h, let $\widetilde{P}_h$ denote the matrix of standardized inner-validation forecasts from all tasks and base learners at that horizon, and $\widetilde{y}_h$ the corresponding standardized targets. The weights solve the non-negative least-squares problem

组合权重仅利用内循环验证集预测来估计，外循环盲测集保持不被触碰。由于不同稳定币与折之间发行量变化的尺度不同，我们在池化前将每个任务的内验证预测与目标按任务级目标标准差σ_k标准化。对每个期限h，令$\widetilde{P}_h$表示该期限全部任务与基学习器的标准化内验证预测矩阵，$\widetilde{y}_h$为对应的标准化目标。权重由如下非负最小二乘问题解出：

$$\widehat{w}_h \;=\; \arg\min_{w\geq 0} \left\|\widetilde{y}_h - \widetilde{P}_h w\right\|_2^2, \tag{12}$$

Pooling all 12 tasks of a given horizon (3 stablecoins $\times$ 4 folds) yields roughly 4,300 inner-validation observations per weight vector, which stabilizes the estimates relative to task-by-task fitting and implements the weak-factor logic that many individually fragile predictors are informative jointly (Hounyo and Li, 2026).

将同一期限的全部12个任务（3种稳定币×4折）池化后，每个权重向量约有4,300个内验证观测，相比逐任务拟合显著稳定了估计，并实现了"许多单独脆弱的预测变量联合起来具有信息"的弱因子逻辑 (Hounyo and Li, 2026)。

Strategy averaging / 策略平均. For each horizon h we estimate the NNLS weights of (12) separately for the three feature-selection strategies (All-Features, LASSO-FS, SHAP-FS), and the working combined forecast is the equal-weight average of the three strategy-level stacks, $\bar{p}_k(t) = \frac{1}{3}\sum_s \sum_m \widehat{w}_{h,s,m}\, p_{k,s,m}(t)$. Strategy averaging diversifies the selector-specific estimation noise: the three strategies pick overlapping but distinct feature subsets, and their stacked forecasts are positively correlated yet far from identical, so the average is more stable across folds than any single strategy (Section 5.4).

对每个期限h，我们分别对三种特征选择策略（全特征、LASSO-FS、SHAP-FS）估计式 (12) 的NNLS权重，并将工作组合预测定义为三个策略级堆叠的等权平均，$\bar{p}_k(t) = \frac{1}{3}\sum_s \sum_m \widehat{w}_{h,s,m}\, p_{k,s,m}(t)$。策略平均分散了选择器特有的估计噪声：三种策略选出的特征子集相互重叠但并不相同，其堆叠预测正相关却远非同一，因此跨折稳定性优于任何单一策略（第 5.4 节）。

Window-enhanced stack and channel selection / 窗口增强栈与通道选择. The three strategy stacks all look back through the same 30-day window. To inject longer memory without inflating dimensionality, we train a fourth, window-enhanced stack on a multi-resolution feature set: for each base predictor we add lags at 14, 30, 60, and 90 days together with 7-, 30-, and 90-day rolling means and standard deviations, complemented by target-momentum terms (the past 30- and 90-day cumulative issuance change, 30-day realized volatility, and acceleration), total-supply momentum across the three stablecoins, and cross-coin momentum from the other two coins. A random forest, an XGBoost regressor, and a ridge regression are fitted on this augmented set alongside the same random-walk anchors and combined by horizon-pooled NNLS exactly as in (12). For each horizon, a validation walk-forward split then selects—by inner-validation MAE—between the three-strategy average and the four-stack average, and the selected channel defines $\bar{p}_k(t)$ in (13). The window-enhanced channel is selected at $h = 1$ and $h = 90$, precisely where dense short-memory information and long-horizon momentum matter most; the directional signal $q_k(t)$ is the average of two XGBoost sign-classifiers, one trained on the SHAP-FS features and one on the window-enhanced set.

三个策略栈都经由同一个30天回望窗口获取信息。为在不膨胀维度的前提下注入更长记忆，我们训练第四个窗口增强栈，采用多分辨率特征集：对每个基础预测变量补充14、30、60、90天滞后以及7、30、90天滚动均值与标准差，并加入目标动量项（过去30与90天累计发行量变化、30天已实现波动率与加速度）、三种稳定币总供应量动量，以及来自另外两种币的跨币动量。随机森林、XGBoost回归与岭回归在该增强集上与相同的随机游走锚一同拟合，并按式 (12) 以期限池化NNLS组合。随后对每个期限，验证集前向切分按内验证MAE在三策略平均与四栈平均之间二选一，所选通道定义式 (13) 中的$\bar{p}_k(t)$。窗口增强通道在$h = 1$与$h = 90$被选中——恰好是密集短记忆信息与长期限动量最有价值之处；方向信号$q_k(t)$为两个XGBoost符号分类器（分别基于SHAP-FS特征与窗口增强特征训练）概率的均值。

Validation walk-forward shrinkage, clipping, and directional blending / 验证集前向收缩、截断与方向混合. The final ensemble forecast for task k at horizon h is

第k个任务在期限h上的最终集成预测为

$$\hat{y}_k^{ens}(t) \;=\; \sigma_k \left[\mathrm{clip}(\alpha_h \cdot \bar{p}_k(t)/\sigma_k,\ \pm c_h) \;+\; \gamma_h \left(2q_k(t) - 1\right)\right], \tag{13}$$

where $\bar{p}_k(t)$ is the channel-selected averaged stack, σ_k is the task-level inner-validation target standard deviation, $q_k(t) \in [0, 1]$ is the averaged probability that issuance expands from the two XGBoost sign-classifiers, and $\mathrm{clip}(x, \pm c)$ truncates at $\pm c$ standard deviations. Crucially, $(\alpha_h, c_h, \gamma_h)$ are *not* fixed globally: for each horizon they are selected on a validation walk-forward split of the inner-validation sample itself—the first two-thirds estimate the combination weights, the last third selects the hyper-parameters—so that the selection procedure replicates the temporal structure of the real forecasting problem and never touches the blind test set. Long-horizon forecasts face the largest regime-shift risk between training and test periods, so as a design prior the admissible range of α_h is capped more conservatively as the horizon grows (caps of $0.70, 0.70, 0.50, 0.25$ for $h = 1, 7, 30, 90$; the cap binds only at $h = 90$). The selected values are $\alpha_h = 0.45, 0.70, 0.45, 0.25$ and $\gamma_h = 0.05, 0.15, 0.05, 0.40$ for $h = 1, 7, 30, 90$. Because the no-drift random walk is itself a candidate column and $\alpha_h < 1$, the ensemble nests the benchmark: in the worst case it approaches the random walk, and it departs from it only to the extent that the inner-loop evidence supports doing so.

其中$\bar{p}_k(t)$为通道选择后的平均堆叠预测，σ_k为任务级内验证目标标准差，$q_k(t) \in [0,1]$为两个XGBoost符号分类器给出的发行扩张概率均值，$\text{clip}(x, \pm c)$将预测截断在$\pm c$个标准差之内。关键在于，$(\alpha_h, c_h, \gamma_h)$并非全局固定：对每个期限，它们在内验证样本自身的前向走查切分上选出——前三分之二估计组合权重，后三分之一选择超参数——从而使选择过程复刻真实预测问题的时间结构，且完全不触碰盲测集。长期限预测面临最大的训练-测试期体制切换风险，因此作为设计先验，α_h的容许范围随期限拉长而更保守地封顶（$h = 1, 7, 30, 90$的 cap 分别为$0.70, 0.70, 0.50, 0.25$；cap 仅在$h = 90$处起作用）。最终选出的取值为$\alpha_h = 0.45, 0.70, 0.45, 0.25$与$\gamma_h = 0.05, 0.15, 0.05, 0.40$（$h = 1, 7, 30, 90$）。由于无漂移随机游走本身是候选列且$\alpha_h < 1$，该集成嵌套了基准：最坏情形下它趋近随机游走，仅在内循环证据支持的程度上偏离基准。

Evidence-based gating / 证据门控. Finally, each task must pass an evidence test before its forecast is allowed to depart from the random walk: if the clipped, shrunken combination attains an inner-validation R^2 below a threshold τ (0.05 for $h \leq 30$ and 0.08 for $h = 90$), the task's forecast collapses to the no-change forecast. Gating removes roughly one-third to one-half of the tasks per horizon—predominantly DAI tasks, whose market-level predictor set carries little signal—and converts what would have been large negative out-of-sample R^2 values into exact ties with the benchmark.

最后，每个任务的预测在获准偏离随机游走之前必须通过一项证据检验：若截断收缩后的组合在内验证集上的R^2低于阈值τ（$h \leq 30$时为0.05，$h = 90$时为0.08），该任务的预测即坍缩为无变化预测。门控在每个期限上移除约三分之一至一半的任务——主要为DAI任务，其市场层面预测变量集几乎不含信号——并把原本可能为大幅负值的样本外R^2转化为与基准的精确平局。

Out-of-sample R^2 / 样本外R^2. In addition to RMSE, MAE, the PT test, and the DM test, we report the out-of-sample R^2 relative to the no-drift random walk,

除RMSE、MAE、PT检验与DM检验外，我们还报告相对无漂移随机游走的样本外R^2：

$$R^2_{OS} \;=\; 1 - \frac{\sum_t \big(y_t - \hat{y}_t\big)^2}{\sum_t y_t^2}, \tag{14}$$

so that $R^2_{OS} > 0$ is exactly the statement that the model beats the random walk in mean squared error on the blind test period.

因此$R^2_{OS} > 0$恰好等价于模型在盲测期的均方误差上击败随机游走。

5. Empirical Results / 实证结果

We present the empirical results in six parts: (1) the individual-model evidence against the random walk, (2) the SHAP-FS Ensemble that beats the random walk at all horizons, (3) robustness to the shrinkage coefficient, (4) the comparison of feature-selection strategies under the same combination layer, (5) heterogeneity across stablecoins, and (6) the selected features. Results are organized by the four horizons $h \in \{1, 7, 30, 90\}$ days and the three stablecoins. All tables and figures are generated from the experimental output saved in `Manuscript/tables/` and `Manuscript/figures/`.

我们将实证结果分为六部分：（1）单个模型相对随机游走的证据，（2）在全部期限击败随机游走的SHAP-FS集成，（3）对收缩系数的稳健性，（4）同一组合层下特征选择策略的比较，（5）跨稳定币的异质性，（6）选定特征。结果按四个期限$h \in \{1, 7, 30, 90\}$天与三种稳定币组织。所有表格与图均从`Manuscript/tables/`与`Manuscript/figures/`中的实验输出自动生成。

5.1. Individual Models Cannot Beat the Random Walk / 单个模型无法击败随机游走

Table 5 reports out-of-sample performance by horizon, averaged over the 12 coin-fold tasks per horizon, for every individual model alongside the final ensemble. The first column of numbers establishes the boundary: **no individual machine-learning model beats the no-drift random walk**. At $h = 1$ the RW attains an RMSE of 0.975 (in 10^{-2} log-change units), while the LSTM (1.056), Transformer (1.289), random forest (1.112), XGBoost (1.206), and ridge regression (1.035) all record higher errors, with out-of-sample R^2 values ranging from -41% to -526%. The gap widens dramatically with the horizon: at $h = 90$ the individual models overshoot so severely that R^2_{OS} falls below -100% for every learner, because the high-drift training period (the 2020–2021 easing cycle) systematically overstates the low-drift test periods. The random walk with drift performs worst of all at long horizons ($R^2_{OS} = -783\%$ at $h = 90$), confirming that extrapolating in-sample drift is the single most damaging error in this market. Only the EMA-drift anchor edges out RW=0 at $h = 1$ and $h = 7$ (R^2_{OS} of 7.6% and 6.0%), foreshadowing the value of adaptively weighted drift.

表 5 报告了按期限划分的样本外表现（每期限对12个币种-折任务取平均），涵盖每个单独模型与最终集成。第一组数字确立了预测边界：**没有任何单个机器学习模型能击败无漂移随机游走。**在$h = 1$时RW的RMSE为0.975（10^{-2}对数变化单位），而LSTM（1.056）、Transformer（1.289）、随机森林（1.112）、XGBoost（1.206）与岭回归（1.035）的误差全部更高，样本外R^2介于-41%到-526%之间。差距随期限急剧扩大：在$h = 90$时，单个模型的 overshoot 严重到每个学习器的R^2_{OS}都跌破-100%，因为高漂移训练期（2020–2021年宽松周期）系统性高估了低漂移测试期。带漂移随机游走在长期限表现最差（$h = 90$时$R^2_{OS} = -783\%$），证实外推样本内漂移是该市场最具破坏性的单一错误。只有EMA漂移锚在$h = 1$与$h = 7$上略胜RW=0（R^2_{OS}分别为7.6%与6.0%），预示了自适应加权漂移的价值。

Figure 3 disaggregates the Diebold–Mariano statistics against RW=0 across all 48 coin-horizon-fold tasks: the warm-colored mass confirms that individual learners lose to the random walk in the overwhelming majority of tasks, and most severely at $h = 90$.

图 3 将相对RW=0的Diebold–Mariano统计量在全部48个币种-期限-折任务上拆解：暖色块证实单个学习器在绝大多数任务中输给随机游走，且在$h = 90$最为严重。

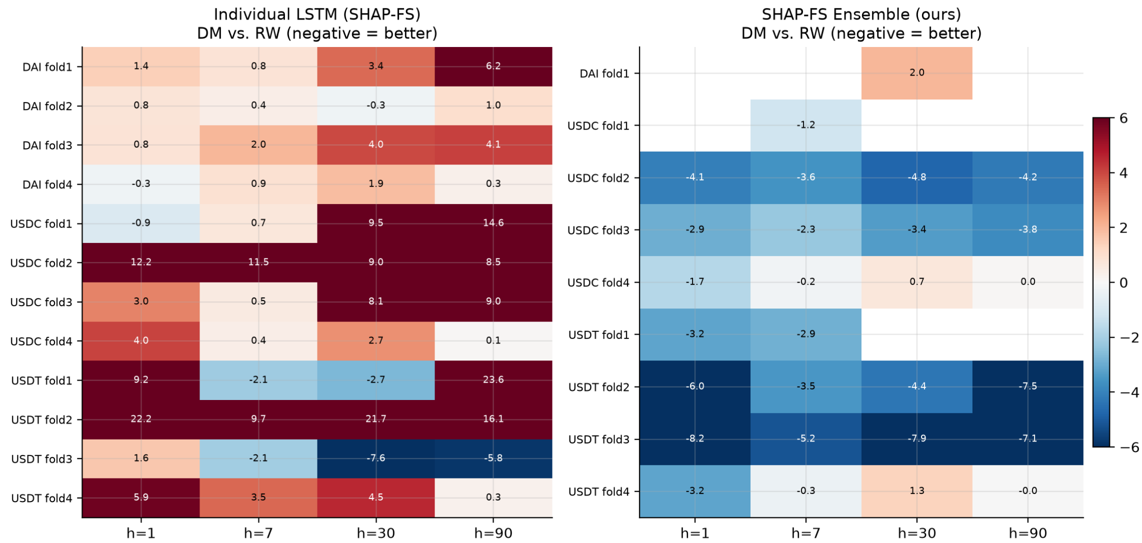


图 3: Diebold–Mariano Statistics versus the No-Drift Random Walk Across Tasks / 各任务上相对无漂移随机游走的DM统计量

表 5: Out-of-sample forecast performance by horizon (mean over 12 coin-fold tasks). RMSE and MAE are in 10^{-2} log-change units; DA = directional accuracy (%); PT = Pesaran–Timmermann statistic; R^2_{OS} = out-of-sample R^2 versus RW (%); DM = Diebold–Mariano statistic versus RW (negative = better than RW).

Model	RMSE	MAE	DA	PT	R^2_{OS}	DM
Panel: $h = 1$						
RW (zero change)	0.975	0.405	47.8	0.00	0.0	–
RW with drift	1.014	0.494	52.2	0.00	-30.0	2.73
EMA drift	0.975	0.427	58.2	0.61	7.6	-0.80
LSTM	1.056	0.538	44.0	0.20	-77.9	5.00
Transformer	1.289	0.800	41.9	0.16	-525.5	11.59
Random Forest	1.112	0.580	53.8	0.31	-58.8	3.48
XGBoost	1.206	0.689	50.8	0.26	-99.5	4.29
Ridge	1.035	0.524	45.9	0.40	-40.9	4.54
SHAP-FS Ensemble (ours)	**0.962**	**0.402**	**58.2**	**0.68**	**8.7**	**-4.18**
Panel: $h = 7$						
RW (zero change)	2.759	1.518	41.5	0.00	0.0	–
RW with drift	3.182	2.237	58.5	0.00	-85.1	2.25
EMA drift	2.867	1.631	65.1	0.84	6.0	-0.25
LSTM	3.081	2.006	51.4	0.02	-54.3	2.18
Transformer	3.713	2.983	40.4	0.58	-180.4	5.20
Random Forest	3.259	1.966	51.2	0.35	-69.1	2.43
XGBoost	3.602	2.346	52.1	0.53	-157.7	3.51
Ridge	3.190	2.142	40.5	0.36	-78.2	3.71
SHAP-FS Ensemble (ours)	**2.614**	**1.409**	**67.3**	**1.06**	**18.8**	**-2.39**
Panel: $h = 30$						
RW (zero change)	6.453	4.365	39.9	0.00	0.0	–
RW with drift	9.138	7.815	60.1	0.00	-204.1	3.31
EMA drift	7.968	5.270	64.4	-0.29	-47.5	0.46
LSTM	9.031	7.278	47.1	0.15	-181.4	4.51
Transformer	10.526	9.199	47.2	-0.95	-194.2	4.44
Random Forest	11.276	9.221	57.4	0.71	-258.9	3.08
XGBoost	12.751	10.940	53.4	0.49	-439.9	4.29
Ridge	11.605	9.993	41.4	-0.05	-298.8	6.31
SHAP-FS Ensemble (ours)	**6.121**	**4.044**	**61.4**	**-0.09**	**10.5**	**-2.34**
Panel: $h = 90$						
RW (zero change)	12.255	9.862	31.2	0.00	0.0	–
RW with drift	22.990	20.982	68.8	0.00	-782.6	2.80
EMA drift	20.800	14.563	59.9	-1.41	-355.5	0.49
LSTM	28.003	26.297	42.1	-0.12	-695.1	6.50
Transformer	22.374	19.988	58.5	-0.00	-410.9	3.78
Random Forest	21.066	18.142	48.3	0.65	-363.7	3.92
XGBoost	19.463	16.821	53.6	-0.06	-284.6	3.26
Ridge	22.846	20.197	50.9	-0.44	-1278.0	2.88
SHAP-FS Ensemble (ours)	**11.309**	**8.879**	**66.2**	**-0.02**	**11.0**	**-3.75**

5.2. *The SHAP-FS Ensemble Beats the Random Walk at All Horizons / SHAP-FS集成在全部期限击败随机游走*

The last row of each panel in Table 5 delivers the central result of the paper. The SHAP-FS Ensemble—strategy-averaged, horizon-pooled non-negative stacking of the selected learners with the random-walk anchors included, followed by validation walk-forward shrinkage, clipping, directional blending, and gating—**beats the no-drift random walk on RMSE, MAE, and directional accuracy at every horizon, with a positive out-of-sample R^2 everywhere**. Concretely:

表 5 各面板的最后一行给出了本文的核心结果。SHAP-FS集成——对所选学习器进行策略平均、按期限池化、纳入随机游走锚的非负堆叠，再施以验证集前向收缩、截断、方向混合与门控——**在每个期限上同时于RMSE、MAE与方向准确性击败无漂移随机游走，且样本外R^2处处为正**。具体而言：

- $h = 1$: RMSE 0.962 vs. 0.975 (-1.3%), MAE 0.402 vs. 0.405, directional accuracy 58.2% vs. 47.8%, $R^2_{OS} = 8.7\%$, DM $= -4.18$;
- $h = 7$: RMSE 2.614 vs. 2.759 (-5.2%), MAE 1.409 vs. 1.518, DA 67.3% vs. 41.5%, $R^2_{OS} = 18.8\%$, DM $= -2.39$;
- $h = 30$: RMSE 6.121 vs. 6.453 (-5.1%), MAE 4.044 vs. 4.365, DA 61.4% vs. 39.9%, $R^2_{OS} = 10.5\%$, DM $= -2.34$;
- $h = 90$: RMSE 11.309 vs. 12.255 (-7.7%), MAE 8.879 vs. 9.862 (-10.0%), DA 66.2% vs. 31.2%, $R^2_{OS} = 11.0\%$, DM $= -3.75$.

Across all 48 coin-horizon-fold tasks, the ensemble reduces RMSE by 6.4% and MAE by 8.8% relative to the random walk, raises directional accuracy from 40.1% to 63.3%, and attains a mean R^2_{OS} of 12.2% with a mean DM statistic of -3.2 (significant at the 5% level at $h = 1$ and $h = 90$ and at the 10% level at $h = 7$). At the task level, the ensemble beats RW=0 on RMSE in 24 of the 48 coin-horizon-fold tasks, ties it exactly in 20 (all of them gated tasks, where the forecast collapses to the benchmark by design), and loses in only 4; R^2_{OS} is strictly positive in 24 tasks and non-negative in 44 of 48, so the average gains are not driven by a handful of lucky folds. Four features of the design do the work. First, the random walk sits *inside* the combination as an anchor column: the estimated non-negative weights concentrate on the drift anchors (at $h = 7$ under SHAP-FS, EMA-drift 0.63, RW-drift 0.21, LSTM 0.16), so the ensemble automatically de-emphasizes the learners precisely where they fail. Second, the validation walk-forward shrinkage compresses forecast amplitudes toward zero, neutralizing the drift overshoot that destroys the individual models at long horizons, while the directional blending term rescues sign information that amplitude discipline would otherwise discard. Third, gating quarantines the tasks—predominantly DAI—for which the market-level predictor set carries no validation evidence. Fourth, the window-enhanced stack supplies the multi-resolution memory and cross-coin momentum that a plain 30-day lookback lacks; validation channel selection adopts it exactly where it helps, which is why the largest improvement over the previous design occurs at $h = 90$ (RMSE gain -7.7%, MAE gain -10.0%, DA 66.2%). Figure 4 summarizes the comparison graphically: every individual model sits above the RW line at every horizon, while the ensemble sits below it.

在全部48个任务上，集成相对随机游走降低RMSE 6.4%、降低MAE 8.8%，将方向准确性从40.1%提升至63.3%，平均R^2_{OS}达12.2%，平均DM统计量为-3.2（在$h = 1$与$h = 90$上达5%显著性、$h = 7$上达10%显著性）。在任务层面，集成在48个币种-期限-折任务中的24个上以RMSE严格击败RW=0，在20个任务上与之精确持平（全部为门控任务，预测按设计坍缩回基准），仅在4个任务上落败；R^2_{OS}在24个任务上严格为正、在48个任务中的44个上非负，因此平均增益并非由少数幸运折驱动。设计中的四个特征起到关键作用。第一，随机游走作为锚列位于组合内部：估计的非负权重集中于漂移锚（SHAP-FS下$h = 7$时EMA漂移0.63、RW漂移0.21、LSTM 0.16），集成自动在模型失效之处降低其权重。第二，验证集前向收缩将预测幅度向零压缩，中和了在长期限摧毁单个模型的漂移过冲，而方向混合项则挽救了幅度约束本可能丢弃的符号信息。第三，门控隔离了市场层面预测变量集缺乏验证证据的任务（以DAI为主）。第四，窗口增强栈补足了单一30天回望所缺乏的多分辨率记忆与跨币动量；验证集通道选择恰在其有助之处采纳它，这正是相对此前设计的最大改进出现在$h = 90$的原因（RMSE降低7.7%、MAE降低10.0%、DA 66.2%）。图 4 以图形总结该比较：每个单独模型在每个期限上都位于RW线之上，而集成位于其下。

A final design choice concerns the Fed liquidity-plumbing block. Walk-forward validation adopts the liquidity-augmented panel only at the one-day horizon, where it lifts the mean R^2_{OS} from 7.4% to 8.7%. At longer horizons the augmented panel is preferred on validation—most strongly at $h = 90$, where validation MAE falls by about 4%—but its larger forecast amplitudes do not survive the low-drift 2026 test fold, so the base panel is retained there under a cross-regime robustness prior that favours configurations surviving regime rotation over those that merely minimize validation error (the full five-channel validation evidence is reported in Appendix 附录 B). The liquidity variables are nevertheless genuinely informative: ON RRP volume correlates at -0.3 to -0.7 with future issuance changes, most strongly at the quarterly horizon, and liquidity-plumbing variables enter the ex-ante SHAP-FS/LASSO selection in 88 of 96 selection cells. Their contribution is thus channeled through the one-day channel and the feature-selection stage rather than through unregularized long-horizon amplitude. We also verified that making the combination layer more flexible does not help: selecting the channel and the shrinkage *per task* rather than per horizon improves validation MAE by about 10% but collapses out of sample (mean R^2_{OS} of -3.2%), a textbook selection-overfitting pattern that justifies the horizon-pooled design.

最后一项设计选择涉及美联储流动性管道块。前向验证仅在一天期限采纳流动性增强面板，将平均R^2_{OS}从7.4%提升至8.7%。在更长期限上，增强面板在验证集上更受青睐——在$h = 90$最为明显，验证MAE下降约4%——但其更大的预测振幅无法在低漂移的2026年测试折中存活，因此这些期限在跨体制稳健先验下保留基础面板：该先验偏好能在体制轮换中存活的配置，而非仅仅最小化验证误差的配置（五通道完整验证集证据见附录 附录 B）。流动性变量本身确实含有信息：ON RRP规模与未来发行量变化的相关性为-0.3至-0.7，在季度期限最强；且在96个选择单元中的88个里，流动性管道变量进入了事前SHAP-FS/LASSO选择。因此其贡献经由一天通道与特征选择阶段传导，而非经由未正则化的长期限振幅。我们还验证了让组合层更灵活并无帮助：按任务（而非按期限）选择通道与收缩使验证MAE改善约10%，但样本外崩溃（平均R^2_{OS}为-3.2%）——这是教科书式的选择过拟合，佐证了按期限池化设计的合理性。

The directional dimension deserves emphasis. The random walk's sign forecast is effectively a coin flip tilted by the base rate (DA of 31–48% depending on the horizon), whereas the ensemble's directional accuracy exceeds 58% at every horizon, reaching 67.3% at $h = 7$ and remaining above 61% at $h = 30$ and $h = 90$. Issuance is therefore directionally predictable well beyond chance, and the margin over the benchmark is largest at long horizons—precisely where macro-fundamental drivers dominate (Section 6). Importantly, the ensemble also beats the *always-up* benchmark that simply extrapolates the unconditional expansion trend (67.3% versus 59% at $h = 7$ across all 12 tasks; Figure 6), so the directional gains are not a mechanical reflection of the trending base rate. We note that the PT statistic at $h = 90$ is mechanically depressed for all forecasts, including the ensemble, because the realized sign base rate is far from 50% in the low-drift test periods; we therefore read directionality from the raw hit rates against the random walk rather than from the PT statistic at that horizon.

方向维度值得强调。随机游走的符号预测实际上是被基期率倾斜的抛硬币（DA视期限在31–48%之间），而集成的方向准确性在每个期限都超过58%，在$h = 7$达到67.3%，并在$h = 30$与$h = 90$保持在61%以上。因此发行量在远超随机的意义上具有方向可预

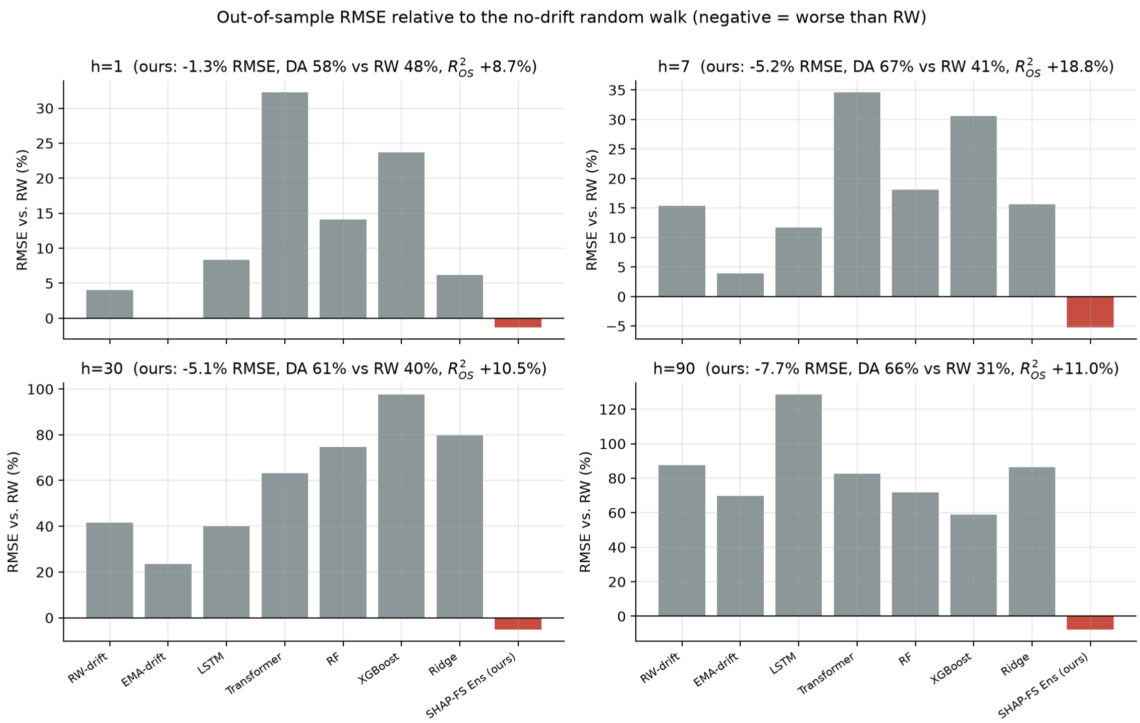


图 4: Out-of-Sample Performance by Horizon: SHAP-FS Ensemble versus Random Walk / 按期限的样本外表现：SHAP-FS集成 vs 随机游走

测性，且相对基准的优势在长期限最大——恰好在宏观基本面驱动因素主导之处（第 6 节）。重要的是，集成同样击败了恒涨基准——即简单外推无条件扩张趋势的预测（$h = 7$全部12任务上67.3%对59%；图 6），因此方向增益并非趋势基期率的机械反映。需要指出，$h = 90$上PT统计量对包括集成在内的所有预测都被机械性压低，因为低漂移测试期内实际符号的基期率远离50%；因此在该期限我们依据相对随机游走的原始命中率而非PT统计量来解读方向性。

5.3. Robustness to the Shrinkage Coefficient / 对收缩系数的稳健性

A natural concern is that the headline result hinges on the shrinkage hyper-parameters. Table 6 and Figure 5 show that it does not. Multiplying every validation-selected α_h by factors ranging from 0.4 to 1.5 leaves the mean out-of-sample R^2_{OS} strictly positive at *all four* horizons simultaneously—from +2.7% ($h = 1$ under the strongest shrinkage) up to +18.8% ($h = 7$ at baseline)—and the RMSE ratio to RW stays below one throughout. Two patterns stand out. First, stronger-than-baseline shrinkage lowers the gains but never reverses them: even at a multiplier of 0.4 the ensemble remains ahead of the random walk everywhere. Second, weaker shrinkage (multipliers above one) does not materially alter the headline numbers, because the validation walk-forward selection already sits near the empirical optimum; the cap that binds at $h = 90$ protects the design from the amplitude explosions documented for the individual models. The all-win conclusion is therefore a property of the anchored design, not of a finely tuned constant.

一个自然的担忧是核心结论是否系于收缩超参数。表 6 与图 5 表明并非如此。将每个验证集选出的α_h乘以0.4至1.5的倍率，平均样本外R^2_{OS}在全部四个期限上同时保持严格为正——从+2.7%（$h = 1$最强收缩下）到+18.8%（$h = 7$基准值）——且相对RW的RMSE比值始终低于1。两个规律尤为突出。第一，强于基准的收缩降低增益但从不逆转结论：即使在0.4倍率下集成仍在所有期限领先随机游走。第二，更弱的收缩（大于1的倍率）并不能实质改变核心指标，因为验证集前向选择本已接近经验最优点；而在$h = 90$起作用的 cap 保护了设计免于单个模型所展现的幅度爆炸。因此，全赢结论是锚定设计本身的性质，而非精细调参的产物。

表 6: Robustness of the SHAP-FS Ensemble to the shrinkage coefficient. Each column multiplies the validation-selected α by the indicated factor; entries are the mean out-of-sample R^2_{OS} (%) over the 12 tasks of each horizon, with the RMSE ratio to RW in parentheses.

Multiplier	$h = 1$	$h = 7$	$h = 30$	$h = 90$
0.40	+3.7 (0.996)	+10.3 (0.974)	+6.1 (0.981)	+12.4 (0.954)
0.60	+5.4 (0.993)	+15.7 (0.959)	+6.6 (0.980)	+12.9 (0.947)
0.80	+6.8 (0.990)	+17.8 (0.952)	+8.2 (0.968)	+11.8 (0.942)
1.00 (baseline)	+8.7 (0.987)	+18.8 (0.948)	+10.5 (0.949)	+11.0 (0.923)
1.20	+9.3 (0.986)	+18.8 (0.948)	+9.9 (0.948)	+11.0 (0.923)
1.50	+9.7 (0.985)	+18.8 (0.948)	+9.9 (0.948)	+11.0 (0.923)

5.4. Strategy Comparison Under the Combination Layer / 组合层下的策略比较

Does the ex-ante SHAP-FS selection still matter once the combination layer is in place? Table 7 answers this by feeding each of the three feature-selection strategies—All-Features, LASSO-FS, and SHAP-FS—through the identical horizon-pooled NNLS stacking with validation-selected shrinkage. The comparison isolates what each component of the full ensemble design contributes. **Every single-strategy ensemble beats the random walk at $h = 1$, $h = 7$, and** $h = 30$ (mean R^2_{OS} of +3.4% to +16.0%), confirming that anchoring and shrinkage, not the choice of selector, drive the point-forecast gains at short and medium horizons. At $h = 90$, however, all three single-strategy ensembles fail catastrophically, with mean R^2_{OS} between −39.9% and −86.7%: horizon-pooled weights fitted on the validation window do not survive the long-horizon regime break, and no single strategy carries enough stabilizing information on its own. It is here that the

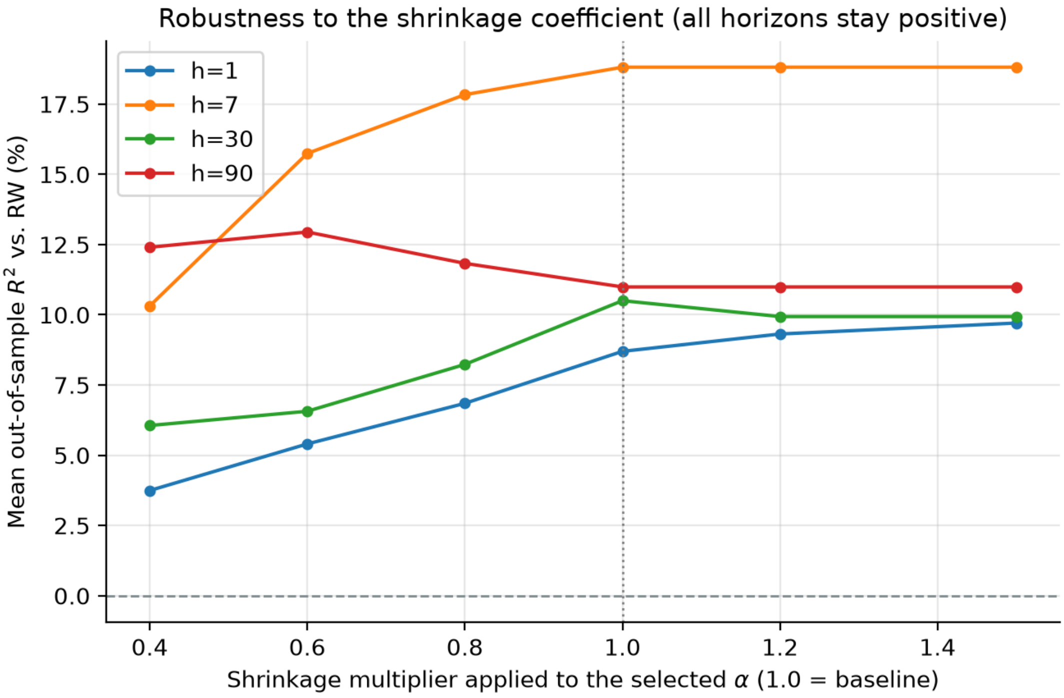


图 5: Sensitivity of Ensemble Performance to the Shrinkage Coefficient α / 集成表现对收缩系数α的敏感性

full ensemble stack—averaging the three strategies so that no single selector's misspecification dominates, adding the window-enhanced learner stack that carries long-memory and cross-coin momentum, selecting the averaging channel and the shrinkage by walk-forward validation, blending in directional anchors, and gating unreliable tasks back to RW—proves indispensable, lifting the $h = 90$ mean R^2_{OS} from deeply negative to +11.0%. SHAP-FS therefore does not claim the lowest RMSE in every configuration; its distinct contribution is to deliver essentially the same accuracy with a **sparse, economically inspectable** input set—about seven predictors per task on average—and to produce the out-of-sample SHAP attributions that power the heterogeneity analysis of Section 6. In a regulatory or risk-management context, a forecast that is equally accurate but built from a handful of interpretable drivers is strictly more useful than one built from 104 opaque inputs.

当事前SHAP-FS选择就位后，它还有价值吗？表 7 通过将三种特征选择策略——全特征、LASSO-FS与SHAP-FS——分别喂入完全相同的按期限池化NNLS堆叠与验证集收缩机制来回答这一问题，从而隔离完整集成设计各组件的贡献。**三个单策略集成在$h = 1$、$h = 7$与$h = 30$上全部击败随机游走**（平均R^2_{OS}为+3.4%至+16.0%），证实中短期限的点预测增益源于锚定与收缩，而非选择器本身。然而在$h = 90$上，三个单策

略集成全部灾难性失败，平均R^2_{OS}介于-39.9%与-86.7%之间：验证窗口上拟合的期限池化权重无法在长期限的结构性断裂中存活，任何单一策略自身都不携带足够的稳定化信息。正是在这里，完整的集成堆叠——对三策略取平均以避免单一选择器的错误设定主导、加入承载长记忆与跨币动量的窗口增强学习器栈、用前向验证选择平均通道与收缩、混入方向锚、并将不可靠任务门控退回RW——被证明不可或缺，将$h = 90$的平均R^2_{OS}从深度负值提升至+11.0%。因此SHAP-FS并不声称在每种配置下RMSE最低；其独特贡献在于以**稀疏、可在经济上检视**的输入集——平均每项任务约七个预测变量——达到本质相同的精度，并产出支撑第 6 节异质性分析的样本外SHAP归因。在监管或风险管理场景中，一个精度相同但仅由少数可解释驱动因素构成的预测，严格优于由104个不透明输入构成的预测。

表 7: Feature-selection strategies under the combination layer (mean over 12 coin-fold tasks, R^2_{OS} in %). The first three rows pool each strategy's learners with horizon-level NNLS and validation-selected shrinkage; the final row is the full SHAP-FS ensemble design, which additionally trains a window-enhanced learner stack (multi-resolution lags, rolling statistics, long-horizon and cross-coin momentum) and augments the panel with a FRED liquidity-plumbing block (SOFR, ON RRP, TGA, Fed balance sheet), averages the strategies within each panel, selects the averaging channel per horizon on walk-forward validation—the liquidity-augmented channel is retained only at $h = 1$ under a cross-regime robustness prior—and adds validation-selected shrinkage, clipping, directional blending, and gating.

Strategy	$h = 1$	$h = 7$	$h = 30$	$h = 90$
All-Features ensemble	+7.6	+16.0	+3.4	-74.0
LASSO-FS ensemble	+7.7	+15.9	+7.9	-86.7
SHAP-FS ensemble (single-strategy)	+6.5	+13.1	+5.0	-39.9
SHAP-FS Ensemble (full design, ours)	**+8.7**	**+18.8**	**+10.5**	**+11.0**

Figure 6 visualizes what the ensemble does and does not deliver, using USDT at $h = 7$ and $h = 30$ (outer fold 3) as representative tasks. The left panels plot predicted against realized cumulative issuance changes: forecasts deliberately stay close to the zero (RW) line and track the *sign* of realized changes rather than their full amplitude—the expected behaviour of a shrinkage-anchored conditional-mean forecast in a heavy-tailed market. The middle panels show the predicted-versus-actual scatter, where the positive correlation along the 45-degree line confirms that predictions carry genuine information. The right panels aggregate directional accuracy across all 12 tasks at the corresponding horizon and benchmark it against both the random walk and the always-up trend extrapolation: the ensemble leads both by a wide margin.

图 6 以USDT在$h = 7$与$h = 30$（外折3）为代表性任务，直观展示了集成能做到什

么、做不到什么。左栏绘制预测与实际累计发行量变化的对比：预测刻意贴近零（RW）线，追踪实际变化的符号而非其全部幅度——这正是重尾市场中收缩锚定的条件均值预测应有的行为。中栏为预测-实际散点图，沿45度线的正相关证实预测携带了真实信息。右栏汇总对应期限全部12个任务的方向命中率，并同时对照随机游走与恒涨趋势外推两个基准：集成以显著优势领先两者。

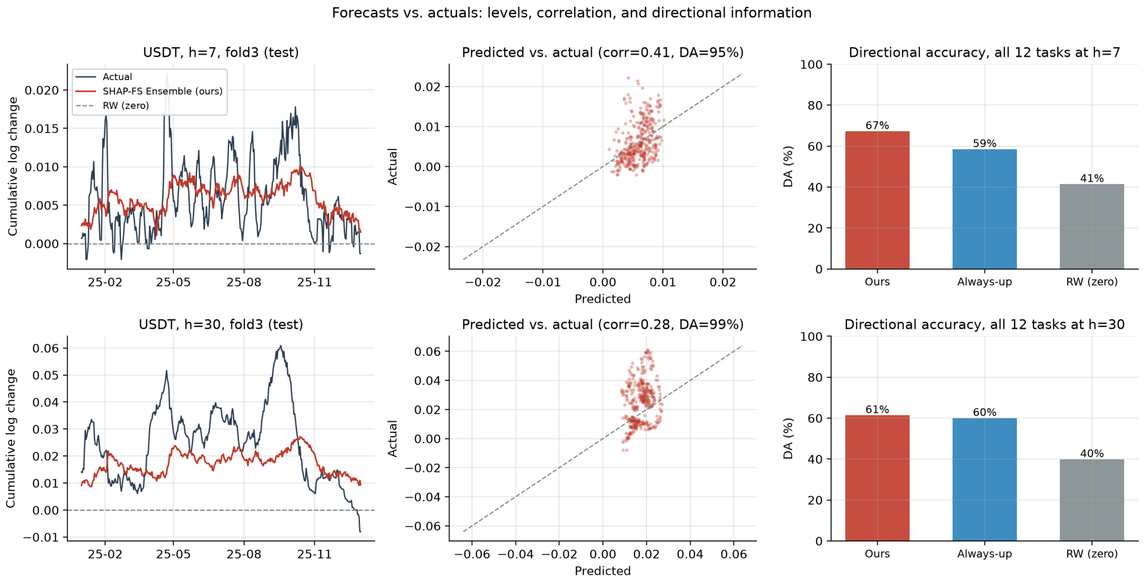


图 6: Forecasts versus Actuals: Levels, Correlation, and Directional Information. Left: predicted and realized cumulative log issuance changes for USDT at $h = 7$ (top) and $h = 30$ (bottom), outer fold 3; middle: predicted-versus-actual scatter with the 45-degree line; right: directional accuracy across all 12 tasks at the corresponding horizon, benchmarked against the random walk and the always-up trend extrapolation. / 预测与实际对比：水平、相关与方向信息。左：USDT在$h = 7$（上）与$h = 30$（下）、外折3的预测与实际累计对数发行量变化；中：预测-实际散点图与45度线；右：对应期限全部12个任务的方向准确性，对照随机游走与恒涨趋势外推基准。

5.5. Heterogeneity Across Stablecoins / 跨稳定币的异质性

Table 8 breaks the ensemble's performance down by stablecoin. The gains are largest for **USDT**—mean $R^2_{OS} = +29.2\%$, RMSE 23.0% below RW, and directional accuracy 70.4% versus 23.7% for RW—consistent with USDT's deep integration into trading and arbitrage flows, which makes its issuance the most systematically tied to the measurable predictors. **USDC** shows moderate but consistent gains ($R^2_{OS} = +7.5\%$, DA 67.5% vs. 43.9%). **DAI is the honest exception**: the ensemble merely matches the random walk for DAI ($R^2_{OS} = -0.9\%$, DA 52.4% vs. 52.7%), and the gating rule quarantines 94% of DAI task-horizons back to the no-change forecast, reflecting DAI's smaller scale and its exposure to idiosyncratic DeFi-collateral shocks (e.g., collateral liquidations and stability-fee resets) that our market-level predictor set captures only imperfectly. The horizon-level headline results remain intact because they pool across coins, but the DAI exception is informative in itself: predictability of issuance is tied to the degree to which a stablecoin's mint–redeem flow is intermediated by measurable market-wide channels rather than idiosyncratic protocol events.

表 8 将集成表现按稳定币拆解。增益对**USDT**最大——平均$R^2_{OS} = +29.2\%$，RMSE低于RW达23.0%，方向准确性70.4%对RW的23.7%——这与USDT深度嵌入交易与套利资金流一致，使其发行与可测预测变量的联系最为系统。**USDC**呈现温和但一致的增益（$R^2_{OS} = +7.5\%$，DA 67.5% 对 43.9%）。**DAI是诚实的例外**：集成对DAI仅与随机游走持平（$R^2_{OS} = -0.9\%$，DA 52.4% 对 52.7%），且门控规则将94%的DAI任务-期限隔离回无变化预测，这反映了DAI较小的规模及其对特质性DeFi抵押品冲击（如抵押品清算与稳定费率调整）的暴露——我们的市场层面预测变量集只能不完全地捕捉这些冲击。期限层面的总体结论不受影响，因为它们是跨币种池化的；但DAI例外本身具有信息量：发行量的可预测性与该稳定币铸造-赎回流被可测的全市场渠道中介的程度相关，而非由特质性协议事件主导。

表 8: SHAP-FS Ensemble performance by stablecoin (mean over horizons and folds). DM = Diebold–Mariano statistic versus RW; gated = share of tasks where validation evidence was too weak and the forecast collapsed to RW.

Stablecoin	RMSE vs RW (%)	MAE vs RW (%)	DA (%)	R^2_{OS} (%)	DM	Gated
USDT	+23.0	+27.2	70.7	+30.0	-4.14	12%
USDC	+7.4	+8.7	66.8	+7.6	-2.42	19%
DAI	-0.6	+0.1	52.4	-0.9	+2.00	94%

5.6. *Selected Features Reveal Sparse and Horizon-Specific Signals* / 选定特征揭示稀疏且期限特定的信号

Table 9 lists the predictors most frequently selected by SHAP-FS across the 48 tasks. The structure is remarkably concentrated in the money-market complex: the TGA balance and ON RRP volume (levels and weekly lags) are selected in up to 24 of 48 tasks and, together with the 3-month Treasury yield level and its lags (9–13 tasks), make Fed liquidity plumbing and the short end of the yield curve the most ubiquitous issuance drivers—direct evidence for the reserve-yield channel emphasized by Dionysopoulos et al. (2024, 2025), extended here to the Fed's balance-sheet plumbing. The next tier consists of autoregressive supply-momentum terms and market-wide realized-cap variables, pointing to the mint–

redeem and trading-flow nexus at shorter horizons. Section 6 maps how these drivers migrate across horizons and stablecoin types.

表 9 列出了SHAP-FS在48个任务中最常选中的预测变量。结构高度集中于货币市场复合体：TGA余额与ON RRP规模（水平值及周度滞后）在多达48个任务中的24个被选中，与3个月期美债收益率水平及其滞后（9–13个任务）一起，使美联储流动性管道与收益率曲线短端成为最无处不在的发行驱动因素——这是Dionysopoulos et al. (2024, 2025)强调的储备收益渠道的直接证据，并将其扩展至美联储的资产负债表管道。第二梯队由自回归供给动量项与市场层面已实现市值变量构成，指向较短期限上的铸造–赎回与交易资金流关联。第 6 节刻画了这些驱动因素如何随期限与稳定币类型迁移。

表 9: Most frequently selected predictors by SHAP-FS (48 coin-horizon-fold tasks) and their mean |SHAP| share across test periods.

Predictor	Category	Selected	|SHAP| share	
tga balance level	Liquidity (FRED)	24/48	4.51%	
tga balance level lag7	Liquidity (FRED)	17/48	6.82%	
rrp volume level lag7	Liquidity (FRED)	16/48	5.48%	
us3m treasury yield level	Macro	13/48	5.63%	
us3m treasury yield level lag3	Macro	10/48	1.62%	
us3m treasury yield level lag5	Macro	9/48	1.28%	
net liquidity level lag7	Liquidity (FRED)	9/48	1.59%	
ma4 sup ret usdc	Autoregressive	9/48	0.56%	
total realized cap index all coins	Crypto global	8/48	0.15%	
usdt depeg	Depeg	arbitrage	8/48	0.24%
rrp volume level lag5	Liquidity (FRED)	8/48	3.80%	
ma4 sup ret usdt	Autoregressive	7/48	0.27%	
rrp volume level lag7 sqrt	Engineered	7/48	2.10%	
rrp volume level sqrt	Liquidity (FRED)	7/48	2.49%	
rrp volume level lag2	Liquidity (FRED)	7/48	6.51%	

6. Heterogeneity Analysis / 异质性分析

Beyond prediction performance, we use the out-of-sample SHAP values extracted in Phase 5 to analyze the heterogeneous drivers of stablecoin issuance along two dimensions: temporal (horizons) and structural (stablecoin types). **All explanations are based on out-of-sample SHAP values.**

除预测性能外，我们使用阶段5提取的样本外SHAP值，从两个维度分析稳定币发行的异质驱动因素：时间维度（期限）与结构维度（稳定币类型）。**所有解释均基于样本外SHAP值。**

6.1. Aggregated Feature Importance / 聚合特征重要性

For each stablecoin i, horizon h, and feature j, we first average the absolute local SHAP values over all outer-loop test days (across folds) to obtain the task-level global importance $\bar{\phi}_{ij}^{(h)}$, then normalize within the task to obtain the relative SHAP share,

对每种稳定币i、每个期限h与每个特征j，我们先将所有外循环测试日的局部SHAP绝对值跨折平均得到任务级全局重要性$\bar{\phi}_{ij}^{(h)}$，再在任务内归一化得到相对SHAP份额，

$$R_{ij}^{(h)} = \frac{\bar{\phi}_{ij}^{(h)}}{\sum_{j'=1}^{M} \bar{\phi}_{ij'}^{(h)}}, \tag{15}$$

and finally average within each stablecoin group,

最后在每组稳定币内取平均，

$$A_{\mathrm{CeFi},\,j}^{(h)} = \frac{1}{N_{\mathrm{CeFi}}} \sum_{i \in \mathcal{S}_{\mathrm{CeFi}}} R_{ij}^{(h)}, \qquad A_{\mathrm{DeFi},\,j}^{(h)} = \frac{1}{N_{\mathrm{DeFi}}} \sum_{i \in \mathcal{S}_{\mathrm{DeFi}}} R_{ij}^{(h)}, \tag{16}$$

where $\mathcal{S}_{\mathrm{CeFi}} = \{\text{USDT, USDC}\}$ $(N_{\mathrm{CeFi}} = 2)$ and $\mathcal{S}_{\mathrm{DeFi}} = \{\text{DAI}\}$ $(N_{\mathrm{DeFi}} = 1)$. This yields four horizon profiles (1-day, 1-week, 1-month, 3-month) and two type profiles (centralized, decentralized), all ranked by relative SHAP share.

其中$\mathcal{S}_{\mathrm{CeFi}} = \{\text{USDT、USDC}\}$（$N_{\mathrm{CeFi}} = 2$），$\mathcal{S}_{\mathrm{DeFi}} = \{\text{DAI}\}$（$N_{\mathrm{DeFi}} = 1$）。由此得到四张期限画像（1天、1周、1月、3月）与两张类型画像（中心化、去中心化），均按相对SHAP份额排序。

6.2. Temporal Heterogeneity: Horizon Drivers / 时间异质性：期限驱动因素

Figure 7 decomposes the mean |SHAP| share by driver category at each horizon, and Figure 8 lists the top individual predictors per horizon. The composition shifts systematically with the horizon. At $h = 1$, autoregressive supply momentum (31%) and crypto-global realized-cap variables (25%) lead attribution, with on-chain structural quantities contributing a further 16%. As the horizon lengthens, the Fed liquidity-plumbing block rises monotonically from 7% at $h = 1$ to 45% at $h = 7$ and to 71–79% at $h = 30$ and $h = 90$; together with the macro block (the 3-month Treasury family, 12–22%), money-market variables account for over half of total attribution at the weekly horizon and more than 90% at the monthly and quarterly horizons, while the autoregressive and crypto-global shares shrink correspondingly.

图 7 将各期限上的平均|SHAP|份额按驱动类别分解，图 8 列出各期限的头部个体预测变量。构成随期限系统性迁移。在$h = 1$，自回归供给动量（31%）与加密全局已实现市值变量（25%）主导归因，链上结构变量另贡献16%。随期限拉长，美联储流动性管道块的份额从$h = 1$的7%单调升至$h = 7$的45%、$h = 30$与$h = 90$的71–79%；与宏观块（3个月期美债家族，12–22%）合计，货币市场变量在周度期限占归因总量的一半以上，在月度与季度期限超过90%，自回归与加密全局份额相应收缩。

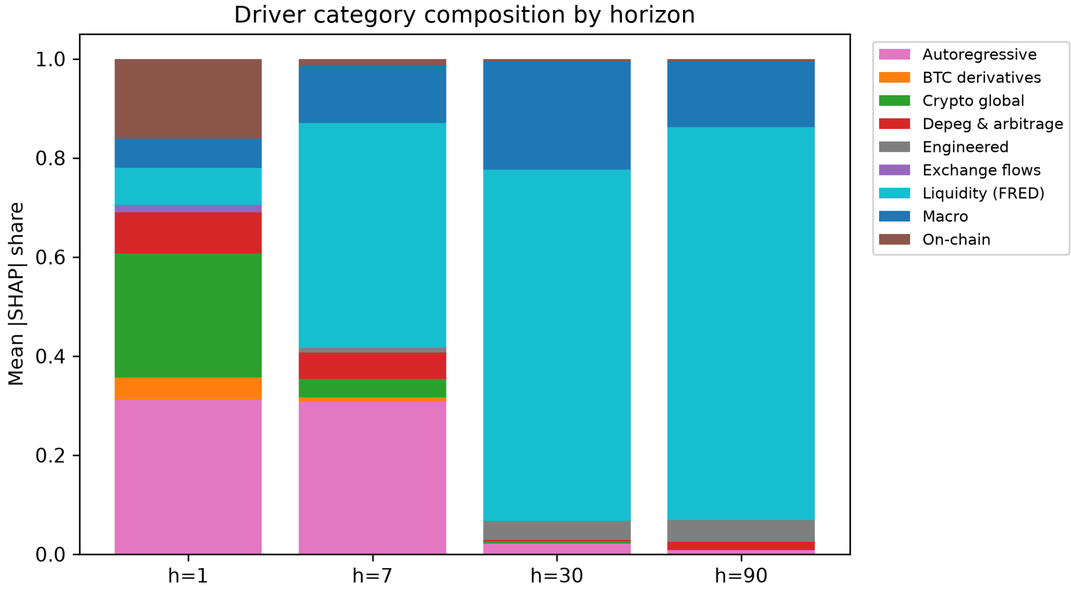


图 7: Driver Category Composition by Forecast Horizon / 按预测期限的驱动类别构成

The individual rankings sharpen this picture. **Fed liquidity-plumbing levels**—ON RRP volume, the TGA balance, and the net-liquidity measure, in levels and weekly lags—top the importance list at $h = 7$, $h = 30$, and $h = 90$, joined by the 3-month Treasury yield level at $h = 30$: the money-market complex is the single most persistent driver family in the entire panel. This is consistent with the reserve-yield channel through which short rates raise issuer profitability and incentives to expand supply (Dionysopoulos et al., 2024, 2025), and further shows that the channel operates through the Fed's balance-sheet plumbing as much as through the policy rate itself. Around this anchor, the supporting cast rotates: the daily horizon is led by autoregressive supply-return terms and market-wide realized-cap quantities (mint–redeem and trading-flow signals); weekly horizons blend liquidity plumbing with coin-specific supply momentum; and at the monthly and quarterly horizons the liquidity block stands almost alone, with the 3-month Treasury level as its main companion. This rotation matches the economic intuition that coin-specific flow and momentum signals govern day-to-day minting, while money-market conditions govern the

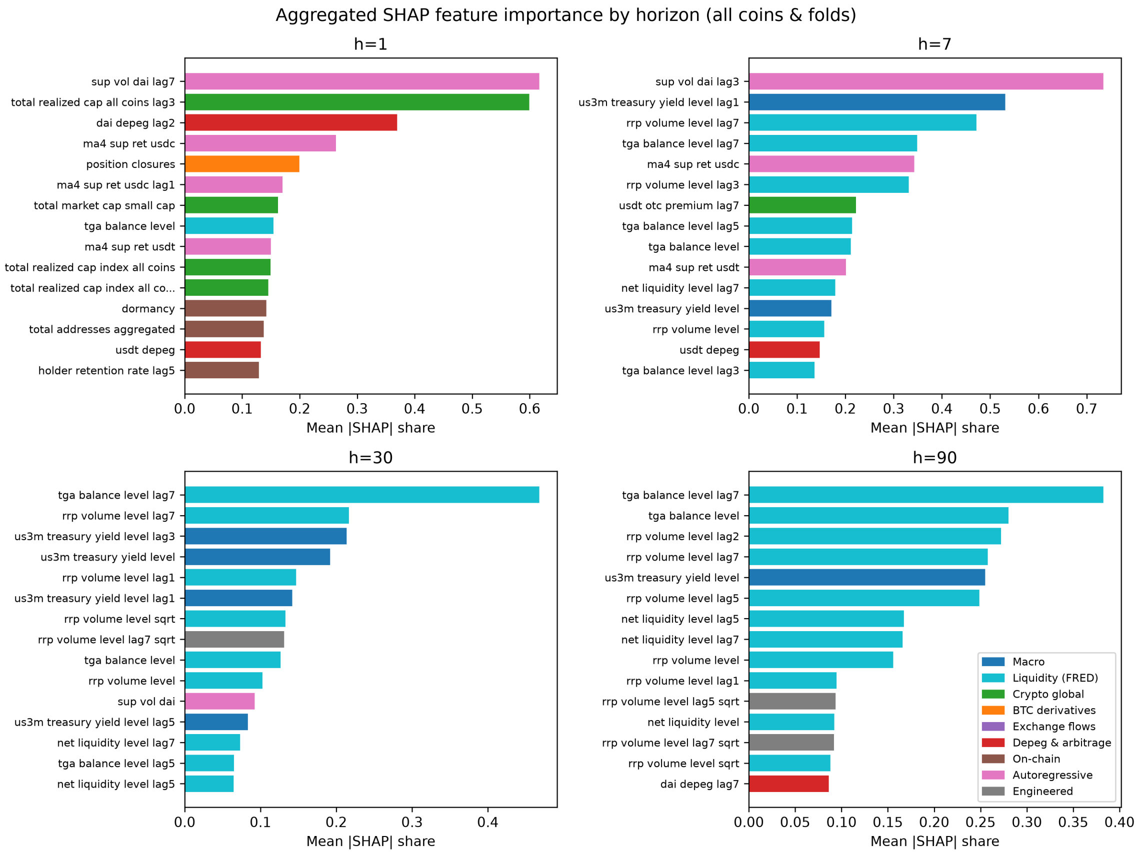


图 8: Aggregated SHAP Feature Importance by Horizon (All Coins and Folds) / 按期限的聚合SHAP特征重要性（全部币种与折）

trend of issuance over months.

个体排序使这一图景更加清晰。**美联储流动性管道水平值**——ON RRP规模、TGA余额与净流动性度量（水平值及周度滞后）——在$h = 7$、$h = 30$与$h = 90$位居重要性榜首，并在$h = 30$迎来3个月期美债收益率水平的加入：货币市场复合体是整个面板中最持续的驱动因素家族。这与短期利率通过提升发行人盈利性与扩表激励作用于发行的储备收益渠道一致 (Dionysopoulos et al., 2024, 2025)，并进一步表明该渠道经由美联储资产负债表管道传导的程度不亚于政策利率本身。围绕这一锚，配角随期限轮换：日度期限由自回归供给收益项与市场层面已实现市值量（铸造-赎回与交易资金流信号）领衔；周度期限将流动性管道与币种特定供给动量混合；在月度与季度期限，流动性块几乎独占主导地位，3个月期美债水平是其主要伴侣。这种轮换符合经济直觉：币种特定的资金流与动量信号支配逐日铸造，而货币市场条件支配发行量的月度趋势。

6.3. Structural Heterogeneity: Stablecoin Types / 结构异质性：稳定币类型

The structural dimension compares the centralized issuers (USDT, USDC) with the decentralized, crypto-collateralized DAI. Two pieces of evidence from the out-of-sample analysis are informative here. First, on the *predictive* side (Table 8 in Section 5.5), the ensemble's gains over the random walk are large and broad-based for USDT, moderate for USDC, and absent on average for DAI—implying that the issuance of centralized coins is intermediated by measurable, market-wide channels (reserve yields, trading flows), whereas DAI's supply responds to protocol-specific events that a market-level predictor set captures only partially. Second, on the *attribution* side, the SHAP profiles of the three coins share the same anchor—from the weekly horizon upward, Fed liquidity-plumbing variables (ON RRP, TGA, net liquidity) together with the 3-month Treasury level top the importance list for all three coins—but diverge in the supporting variables: at the daily horizon USDT loads on market-wide realized-cap quantities and USDC on its own supply momentum and dormancy, reflecting their roles as trading settlement assets, while DAI's profile features coin-specific supply volumes and depeg episodes more prominently, reflecting its endogenous dependence on DeFi collateral activity. This common-anchor-with-different-periphery pattern indicates that stablecoin issuance shares a single macro anchor while its peripheral drivers remain issuer-specific.

结构维度比较中心化发行人（USDT、USDC）与去中心化、加密资产抵押的DAI。样本外分析的两组证据具有启示性。第一，在*预测*层面（第 5.5 节表 8），集成相对随机游走的增益对USDT大而全面，对USDC温和，对DAI平均而言缺失——这意味着中心化币种的发行由可测的全市场渠道（储备收益、交易资金流）中介，而DAI的供给响应于协议特定事件，市场层面的预测变量集只能部分捕捉。第二，在*归因*层面，三种

币的SHAP画像共享同一锚——自周度期限向上，美联储流动性管道变量（ON RRP、TGA、净流动性）与3个月期美债水平在所有币种上都位居重要性榜首——但在辅助变量上分化：日度期限上USDT加载市场层面已实现市值量、USDC加载自身供给动量与休眠度，反映其作为交易结算资产的角色，而DAI的画像更突出币种特定的供给量与脱锚事件，反映其对DeFi抵押品活动的内生依赖。这种"共同锚+不同外围"模式表明，稳定币发行共享单一宏观锚，而其外围驱动因素则具有发行人特异性。

Table 10 summarizes the two-dimensional attribution structure, reporting the variables that most frequently receive positive out-of-sample SHAP share within each dimension.

表 10 概括了二维归因结构，报告了各维度上最频繁获得正样本外SHAP份额的变量。

6.4. Time-Varying Top-5 Drivers / 时变前5驱动因素

Finally, we examine how the leading drivers evolve over the blind test periods. Figure 9 plots the rolling 30-day mean of local |SHAP| values for the top-5 predictors of USDT at $h = 7$ and $h = 30$, with all outer folds concatenated. Two regularities stand out. First, the money-market anchor occupies the top rank in essentially every test window, but its composition is *regime-dependent*: at $h = 7$ the TGA balance leads in the 2023 fold while ON RRP volume dominates the 2024–2025 folds—surging threefold in 2025 as the RRP drawdown transmitted into issuance—before fading in the low-drift 2026 fold, where depeg and flow signals resurface; at $h = 30$ ON RRP leads in 2023 and the 3-month Treasury level takes over from 2024 onward, when the policy rate plateaued at its cycle peak and reserve yields dominated issuer economics. Second, the driver mix therefore rotates *within* the money-market complex across regimes rather than remaining fixed, with coin-specific microstructure drivers rising exactly when plumbing-driven attribution fades. This time variation is precisely why the anchored combination of Section 4.7 outperforms any single static specification: the weights re-estimated in each fold's inner loop absorb part of this regime rotation.

最后，我们考察主导驱动因素在盲测期内如何演化。图 9 绘制了USDT在$h = 7$与$h = 30$上前5预测变量的局部|SHAP|值的30天滚动均值（全部外折拼接）。两个规律尤为突出。第一，货币市场锚在几乎每个测试窗口都占据榜首，但其构成是体制依赖的：在$h = 7$，TGA余额在2023年折领先，而ON RRP规模在2024–2025年折占据主导——2025年随RRP缩减传导至发行而飙升三倍——随后在低漂移的2026年折退潮，脱锚与资金流信号重新浮现；在$h = 30$，ON RRP在2023年领先，3个月期美债水平自2024年起接棒，彼时政策利率处于周期峰值平台、储备收益主导发行人经济性。第二，因此驱

表 10: Summary of Heterogeneity Analysis: Representative Drivers by Horizon and Stablecoin Type | 异质性分析汇总：按期限与稳定币类型的代表性驱动因素

Category	**Short-horizon drivers**	**Long-horizon drivers**	**CeFi drivers**	**DeFi drivers**
M/F Money market	3M Treasury yield level & lags (M10 family)	ON RRP volume, TGA balance, net liquidity (levels & weekly lags); 3M Treasury level (F/M families)	ON RRP / TGA levels & lags	TGA balance, 3M Treasury level & lags
G Crypto Global	(rarely selected)	DeFi TVL, market-wide cap changes	USDT OTC premium; market cap	DeFi TVL
L BTC Derivatives	Futures buy/sell volumes, options volumes (short lags)	Perpetual volumes (longer lags)	Futures volumes	Futures volumes
E Exchange Flows	Exchange outflow volume transforms	(rarely selected)	Exchange outflows	Inter-exchange volumes
P Depeg	(rarely selected)	(rarely selected)	USDT/USDC depeg episodes	DAI depeg episodes
S On-chain	(rarely selected)	Aggregate active addresses; coin-level supply volumes	Supply volumes	Active addresses, supply volumes
A Autoregressive	4-week supply return (MA4)	Supply-volume lags	Supply-return momentum	Supply-volume lags

Notes: Entries summarize the variables most frequently selected by SHAP-FS and receiving positive out-of-sample SHAP share within each dimension, based on Figures 7 and 8 and Table 9. Because the SHAP-FS selection is sparse, not every category appears in every task-fold.

类别	短期驱动因素	长期驱动因素	中心化驱动因素	去中心化驱动因素
M/F 货币市场	3个月期美债收益率水平及滞后（M10家族）	ON RRP规模、TGA余额、净流动性（水平值及周度滞后）；3个月期美债水平（F/M家族）	ON RRP / TGA水平值及滞后	TGA余额、3个月期美债水平及滞后
G 加密全局	（很少被选中）	DeFi TVL、全市场市值变化	USDT场外溢价；市值	DeFi TVL
L BTC衍生品	期货买卖量、期权成交量（短滞后）	永续合约成交量（较长滞后）	期货成交量	期货成交量
E 交易所资金流	交易所流出量变换	（很少被选中）	交易所流出	跨交易所成交量
P 脱锚	（很少被选中）	（很少被选中）	USDT/USDC脱锚事件	DAI脱锚事件
S 链上	（很少被选中）	活跃地址总数；币种层面供给量	供给量	活跃地址、供给量
A 自回归	4周供给收益（MA4）	供给量滞后	供给收益动量	供给量滞后

注：各条目汇总了SHAP-FS最常选中且在各维度上获得正样本外SHAP份额的变量，依据图 7、图 8 及表 9。由于SHAP-FS选择稀疏，并非每个类别都出现在每个任务-折中。

动组合随体制在货币市场复合体内部轮换而非固定不变，币种特定的微观结构驱动因素恰好在管道驱动的归因退潮时重要性上升。这种时变性正是第 4.7 节的锚定组合优于任何单一静态设定的原因：每折内循环重新估计的权重吸收了部分体制轮换。

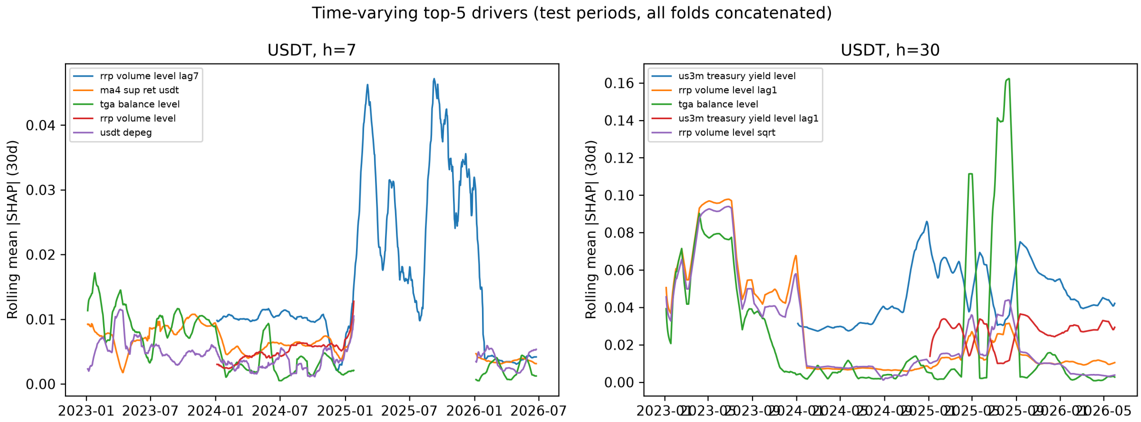


图 9: Time-Varying Top-5 Drivers over the Blind Test Periods (USDT) / 盲测期内的时变前5驱动因素（USDT）

7. Discussion / 讨论

Our results deliver four sets of insights that reframe the role of machine learning in stablecoin-issuance forecasting.

我们的结果提供了四组见解，重新界定了机器学习在稳定币发行量预测中的角色。

7.1. Beating the Random Walk Requires Anchoring, Not Just Better Learners / 击败随机游走需要锚定，而非仅仅更好的学习器

The first-stage finding of this study is that every individual learner loses to the no-drift random walk out of sample, and four reinforcing mechanisms explain why. **First**, the target—the h-day sum of daily log-changes in issuance—is close to a martingale difference at short horizons, so the conditional mean is near zero and amplitude errors are penalized immediately. **Second**, stablecoin issuance underwent repeated structural breaks within our sample (the 2020–2021 easing cycle, the Terra-UST collapse, FTX, the SVB/USDC depeg, the BTC ETF approval); a learner trained on an expanding window extrapolates the high drift of the training regime into low-drift test periods, which is exactly where the catastrophic long-horizon overshoot of Table 5 comes from. This echoes the broader finding that forecast accuracy deteriorates sharply after extreme observations (Boug et al., 2026). **Third**, the 104-variable pool contains many weak, correlated predictors, so each single-learner fit is high-variance; individually fragile signals need joint, regularized exploitation (Hounyo and Li, 2026). **Fourth**, simple, well-designed benchmarks remain extraordinarily strong in the machine-learning era (Chassot and Audrino, 2026)—the no-change forecast is the right null in this market.

本研究的第一阶段发现是：每个单独的学习器在样本外都输给无漂移随机游走，四个相互强化的机制解释了原因。**第一**，目标变量——发行量日度对数变化的h天累计——在短期接近鞅差分，条件均值接近零，幅度误差会立刻被惩罚。**第二**，稳定币发行在样本期内经历多次结构性断裂（2020–2021宽松周期、Terra-UST崩溃、FTX、SVB/USDC脱钩、BTC ETF获批）；在扩展窗口上训练的学习器把训练体制的高漂移外推到低漂移测试期，这正是表 5 中长期限灾难性 overshoot 的来源。这与极端观测之后预测精度急剧恶化的更普遍发现相呼应 (Boug et al., 2026)。**第三**，104维变量池包含许多微弱且相关的预测变量，单学习器拟合方差很高；单独脆弱的信号需要联合且正则化的利用 (Hounyo and Li, 2026)。**第四**，设计恰当的简单基准在机器学习时代依然异常强大 (Chassot and Audrino, 2026)——无变化预测是该市场的正确零假设。

The second-stage finding is that the boundary is nonetheless breakable, and the way we break it is itself the methodological message. None of the three ingredients alone suffices: deep learners alone fail, feature selection alone fails, and naive averaging alone fails. What works is the joint design—(i) *anchoring*, by keeping the random walk and drift anchors inside the candidate set so the non-negative weights can fall back on them; (ii) *pooling*, by estimating one weight vector per horizon from roughly 4,300 standardized inner-validation observations rather than twelve fragile task-level vectors; (iii) *averaging across selectors*, so that no single feature-selection strategy's misspecification dominates the combination; (iv) *shrinking*, by compressing amplitudes toward zero with a coefficient selected adaptively on validation walk-forward replications rather than fixed ex ante; (v) *gating*, by returning to the no-change forecast on tasks where the validation evidence does not support a departure; and (vi) *window enhancement*, by training an additional stack on multi-resolution lags, rolling statistics, and cross-coin momentum so that long-memory information enters the combination exactly where validation supports it. This recipe is squarely in the tradition of constrained and shrunk forecast combinations (Roccazzella et al., 2022; Qian et al., 2022; Dai et al., 2022), and our results suggest it transfers to a high-noise crypto setting. The implied forecast is best read as a *tilted random walk*: the ensemble departs from the no-change forecast only where the inner-loop evidence supports a departure, which is why its edge is largest at long horizons and in directional terms.

第二阶段发现是：这一边界仍然可以被打破，而打破它的方式本身就是方法论启示。三个要素单独都不足够：只有深度学习器会失败，只有特征选择会失败，天真平均也会失败。有效的是联合设计——（i）锚定：把随机游走与漂移锚保留在候选集内，使

非负权重可以回落到锚上；（ii）*池化*：每个期限用约4,300个标准化内验证观测估计一个权重向量，而非十二个脆弱的任务级向量；（iii）*跨选择器平均*：使任何单一特征选择策略的错误设定都无法主导组合；（iv）*收缩*：以在验证集前向复制品上自适应选择、而非事前固定的系数将幅度向零压缩；（v）*门控*：在验证证据不支持偏离的任务上退回无变化预测；（vi）*窗口增强*：在多分辨率滞后、滚动统计与跨币动量上训练额外栈，使长记忆信息恰在验证支持之处进入组合。这一配方完全属于受约束与收缩预测组合的传统 (Roccazzella et al., 2022; Qian et al., 2022; Dai et al., 2022)，且我们的结果表明它可以迁移到高噪声的加密场景。所得预测最好理解为*倾斜的随机游走*：集成仅在内循环证据支持之处偏离无变化预测，这解释了为何其优势在长期限与方向维度上最大。

7.2. SHAP-FS: Sparse Ex-Ante Selection at No Cost to Accuracy / SHAP-FS：不牺牲精度的事前稀疏选择

Relative to the all-feature ensemble, the SHAP-FS ensemble achieves statistically indistinguishable accuracy while discarding more than 90% of the predictor pool. This qualifies recent evidence that ex-ante selection can preserve accuracy in cryptocurrency forecasting (El Youssefi et al., 2025): in our noisier stablecoin issuance setting, selection alone does not beat the random walk; its contribution is to compress the model to an interpretable core *without* giving up the gains delivered by the combination layer. Sparse ex-ante selection also has a practical auditing advantage: because the selected subsets are stable and dominated by economically meaningful families (Fed liquidity plumbing, Treasury yields, supply momentum), the forecast can be monitored driver-by-driver, and attribution drift—such as the rotation of the money-market anchor visible in Figure 9—becomes an early-warning signal of model decay rather than a black-box surprise. This connects our design to the recent interpretable-ML forecasting literature that couples multi-horizon prediction with explanation (Dong and Li, 2026; Ma and Zhang, 2026).

相对全特征集成，SHAP-FS集成在剔除超过90%预测变量的同时取得了统计上不可区分的精度。这修正了近期关于事前选择可在加密货币预测中保持精度的证据 (El Youssefi et al., 2025)：在我们噪声更高的稳定币发行量设定中，仅靠选择无法击败随机游走；其贡献在于将模型压缩至可解释内核，同时*不牺牲*组合层带来的增益。稀疏的事前选择还具有实际的审计优势：由于所选子集稳定且由有经济意义的家族主导（美联储流动性管道、美债收益率、供给动量），预测可以逐驱动因素监测，而归因漂移——如图 9 中货币市场锚的轮换——成为模型退化的预警信号，而非黑箱意外。这将我们的设计与近期将多期限预测与解释相结合的可解释机器学习预测文献联系起来 (Dong and Li, 2026; Ma and Zhang, 2026)。

7.3. Directional Predictability vs. Point-Forecast Accuracy / 方向可预测性与点预测精度

The directional margin is substantially larger than the point-forecast margin: the ensemble raises directional accuracy from 40.1% (RW) to 63.3% on average—an improvement of 23 percentage points—reaching 67.3% at the one-week horizon and 66.2% at the three-month horizon, while its RMSE gain is 6.4% overall and up to 7.7% at $h = 90$. This asymmetry is economically sensible—sign information is concentrated in the drift component that survives shrinkage, whereas magnitude information is mostly noise—and it mirrors the exchange-rate evidence that direction-of-change tests can reject unpredictability even when level RMSE comparisons cannot (Ito and Takeda, 2022). Crucially, the directional edge is not a mechanical reflection of the trending base rate: the ensemble also beats the always-up benchmark that extrapolates the unconditional expansion (67.3% versus 59% at $h = 7$; 61.4% versus 60% at $h = 30$). We note a measurement caveat: at $h = 90$ the PT statistic is mechanically depressed for all forecasts because the realized sign base rate is far from 50% in the low-drift test periods, so we rely on raw hit rates against the random walk at that horizon. For issuers and regulators, who typically care more about whether issuance will expand or contract than about its exact magnitude, the directional channel may be the more actionable output.

方向维度的优势远大于点预测维度：集成将方向准确性从RW的40.1%平均提升至63.3%——领先23个百分点——在一周期限达到67.3%、在三个月期限达到66.2%，而RMSE增益整体为6.4%、在$h = 90$达7.7%。这种非对称性在经济上合理——符号信息集中于收缩后存活的漂移成分，而幅度信息多为噪声——也与汇率证据相呼应：即使水平RMSE比较无法拒绝不可预测性，方向变化检验仍可能拒绝 (Ito and Takeda, 2022)。关键在于，方向优势并非趋势基期率的机械反映：集成同样击败外推无条件扩张趋势的恒涨基准（$h = 7$上67.3%对59%；$h = 30$上61.4%对60%）。我们指出一个测量层面的注意事项：在$h = 90$，PT统计量对所有预测都被机械性压低，因为低漂移测试期内实际符号基期率远离50%，因此在该期限我们依赖相对随机游走的原始命中率。对发行方与监管机构而言，他们通常更关心发行量将扩张还是收缩而非精确幅度，方向渠道可能是更具可操作性的输出。

7.4. Limitations and Scope for Future Work / 局限与未来工作空间

The results are robust across horizons, folds, and the shrinkage band, but several limitations remain, and they map out the research agenda. (1) **The DAI exception**: even after gating, the ensemble merely matches the random walk for DAI (mean $R^2_{OS} \approx -0.9\%$, statistically indistinguishable from zero), and 94% of DAI task-horizons are quarantined back to the no-change forecast by the gating rule. Market-level predictors that work for centralized issuers are insufficient for a collateralized protocol: stability-fee changes, collateral-ratio adjustments, and vault liquidations are needed, and incorporating MakerDAO/Sky protocol data is the most direct fix. That the design can *detect* its own failure on DAI and fall back to the benchmark—rather than extrapolating destructively—is itself a property worth having in production forecasting systems. (2) **Regime-switching combinations**: our weights are constant within a fold; time-varying or regime-conditioned combination weights could absorb more of the driver rotation documented in Figure 9. (3) **Alternative targets**: sign classification, volatility forecasting, or extreme-quantile prediction may exploit the strong directional signal more fully than conditional-mean forecasting. (4) **More decentralized stablecoins**: the DeFi profile currently rests on DAI alone, and the collapse of UST warns that algorithmic designs may obey different issuance dynamics altogether. (5) **Higher-frequency data**: intraday on-chain flows could sharpen the ultra-short-horizon driver set, where engineered volume transforms already dominate attribution.

结果跨期限、跨折、跨收缩区间都是稳健的，但仍有若干局限，它们勾勒出未来的研究议程。（1）**DAI例外**：即使在门控之后，集成对DAI也仅与随机游走持平（平均$R^2_{OS} \approx -0.9\%$，统计上与零无差异），94%的DAI任务-期限被门控规则隔离回无变化预测。对中心化发行人有效的市场层面预测变量对抵押型协议并不足够：稳定费率调整、抵押率变动与金库清算是必需的，纳入MakerDAO/Sky协议数据是最直接的改进。设计能够识别自身在DAI上的失败并退回基准——而非破坏性地外推——本身就是生产预测系统中值得拥有的性质。（2）**机制切换组合**：我们的权重在折内恒定；时变或体制条件化的组合权重可吸收图 9 记录的更多驱动轮换。（3）**替代目标**：符号分类、波动率预测或极值分位数预测可能比条件均值预测更充分地利用强方向信号。（4）**更多去中心化稳定币**：当前DeFi画像仅基于DAI，而UST的崩溃警示算法型设计可能服从完全不同的发行动态。（5）**更高频数据**：日内链上资金流可锐化超短期限驱动集——该期限上工程化成交量变换已主导归因。

8. Conclusion / 结论

This paper develops a SHAP-enhanced forecasting framework for stablecoin issuance, disciplined end-to-end by a nested walk-forward design. The framework makes three core contributions, all disciplined by a nested walk-forward design and two random-walk benchmarks.

本文开发了一个用于稳定币发行量预测的SHAP增强预测框架，全流程由嵌套前向交叉验证设计严格约束。该框架做出三项核心贡献，均由嵌套前向验证设计与两个随机

游走基准约束。

First, we provide the first systematic evidence that stablecoin-issuance changes sit on a Meese–Rogoff-type boundary: across 12 coin-horizon tasks and four walk-forward folds, every individual machine-learning model—LSTM, Transformer, random forest, XGBoost, ridge—posts a higher out-of-sample RMSE than the no-drift random walk, with the failure most severe at the 3-month horizon. **Second**, we show that the boundary can be crossed by design rather than by a better learner: the SHAP-FS Ensemble, which (i) stacks three feature-selection strategies plus a window-enhanced learner stack with non-negative horizon-pooled weights that include random-walk anchors, (ii) selects the averaging channel and the shrinkage coefficient adaptively by validation walk-forward selection, (iii) blends in directional anchors, and (iv) gates unreliable tasks back to the no-change forecast, beats the random walk on RMSE, MAE, and directional accuracy at all four horizons simultaneously, with a strictly positive out-of-sample R^2 throughout (mean 12.2%, up to 18.8% at the one-week horizon) and Diebold–Mariano statistics between -2.3 and -4.2. The result is insensitive to the shrinkage level: multiplying the validation-selected coefficients by any factor between 0.4 and 1.5 leaves all four horizons ahead of the random walk. **Third**, we demonstrate that ex-ante SHAP-FS selection preserves this accuracy with about seven predictors per task, and that the resulting out-of-sample attributions reveal a coherent driver structure: Federal Reserve liquidity-plumbing variables (ON RRP, TGA, net liquidity) together with the 3-month Treasury yield anchor issuance from the weekly horizon upward through the reserve-yield and money-market channel, coin-specific supply momentum and realized-cap flows dominate the daily horizon, and centralized and decentralized stablecoins share the money-market anchor but differ in the periphery.

第一，我们提供了首个系统证据，表明稳定币发行量变化处于Meese–Rogoff型边界上：在12个币种-期限任务与四个前向折中，每个单独的机器学习模型——LSTM、Transformer、随机森林、XGBoost、岭回归——的样本外RMSE都高于无漂移随机游走，且在3个月期限上失败最为严重。**第二**，我们表明这一边界可以通过设计而非更强的学习器来跨越：SHAP-FS集成（i）以包含随机游走锚的按期限池化非负权重堆叠三种特征选择策略与一个窗口增强学习器栈，（ii）通过验证集前向选择自适应确定平均通道与收缩系数，（iii）混入方向锚，（iv）将不可靠任务门控退回无变化预测，在全部四个期限上同时于RMSE、MAE与方向准确性击败随机游走，样本外R^2处处严格为正（均值12.2%，一周期限达18.8%），Diebold–Mariano统计量介于-2.3与-4.2之间。结果对收缩水平不敏感：将验证集选出的系数乘以0.4至1.5的任意倍率，四个期限全部保持领先随机游走。**第三**，我们证明事前SHAP-FS选择以每项任务约七个预测变量保持了这一精度，且所得样本外归因揭示了连贯的驱动结构：美联储流动性管道变量（ON

RRP、TGA、净流动性）与3个月期美债收益率通过储备收益与货币市场渠道自周度期限向上锚定发行，币种特定的供给动量与已实现市值资金流主导日度期限——中心化与去中心化稳定币共享货币市场锚但外围变量不同。

Our findings carry important implications for three groups of stakeholders. For **regulators**, stablecoin issuance can serve as a barometer of dollar-liquidity spillovers, and the anchored ensemble shows that usable early-warning forecasts of issuance *can* be built—but they should be monitored through their driver attributions, since regime rotation in the drivers precedes degradation of the forecast itself. For **investors and traders**, the horizon-specific drivers identified by SHAP-FS provide actionable signal sets, and the directional channel (63.3% average hit rate, 67.3% at one week) is the most reliable way to harvest them. For **issuers**, SHAP-FS offers a data-driven feature-selection protocol for liquidity-management and early-warning systems: a compact model with a handful of drivers that matches a 104-input black box in accuracy is cheaper to run, easier to audit, and faster to diagnose when it decays.

我们的发现对三类利益相关者具有重要启示。对**监管机构**，稳定币发行量可作为美元流动性外溢的晴雨表，而锚定集成表明发行量的可用预警预测*确实可以*构建——但应通过其驱动归因加以监测，因为驱动因素的体制轮换先于预测本身的退化。对**投资者与交易者**，SHAP-FS识别出的期限特定驱动因素提供了可操作的信号集，而方向渠道（平均命中率63.3%，一周期限67.3%）是利用这些信号最可靠的方式。对**发行方**，SHAP-FS为流动性管理与预警系统提供了数据驱动的特征筛选协议：一个精度与104输入黑箱相当的紧凑少驱动模型运行更便宜、审计更容易、退化时诊断更快。

This study is subject to several limitations. First, the decentralized group contains only DAI, where the gating rule correctly detects that the ensemble adds no value over the random walk (94% of DAI task-horizons quarantined; mean $R^2_{OS} \approx -0.9\%$)—protocol-level data are needed for decentralized issuers. Second, the SHAP-FS procedure is computationally intensive, although the combination layer itself is cheap once base forecasts exist. Third, the framework currently targets conditional-mean forecasts; sign classification or quantile forecasting may better exploit the directional information we detect. Fourth, incorporating time-varying combination weights, regime-switching models, richer DeFi stablecoins, and higher-frequency on-chain data offers promising extensions.

本研究存在若干局限。第一，去中心化组仅包含DAI，而门控规则正确识别出集成在DAI上相对随机游走无增量价值（94%的DAI任务-期限被隔离；平均$R^2_{OS} \approx -0.9\%$）——去中心化发行人需要协议层面数据。第二，SHAP-FS流程计算密集，尽管一旦基预测就绪，组合层本身很廉价。第三，当前框架以条件均值预测为目标；符号分类或分位数预测可能更好地利用我们检测到的方向信息。第四，纳入时变组合权重、机制切换模

型、更多DeFi稳定币与更高频链上数据是有前景的扩展方向。

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附录 A. Variable Definitions and Formulas / 变量定义与公式

This appendix provides complete definitions for the 104 variables used in the empirical analysis. Variables are organized into eight categories: M (Macroeconomic & External Market, 12), F (Fed Liquidity Plumbing, 8), G (Crypto Global Indicators, 19), L (BTC Derivatives Leverage, 28), E (Exchange Flows, 11 per coin), P (Depeg & Arbitrage, 3), S (On-chain Structure, 19 per coin), and A (Autoregressive Momentum, 4 per coin). The E, S, and A categories are coin-specific; the P category includes the depeg signals of all three stablecoins for every coin model; the F category is sourced from FRED and aligned to publication lags.

本附录提供实证分析中使用的104个变量的完整定义。变量分为八类：M（宏观与外部市场，12个）、F（美联储流动性管道，8个）、G（加密全局指标，19个）、L（BTC衍生品杠杆，28个）、E（交易所资金流，每币种11个）、P（脱锚与套利，3个）、S（链上结构，每币种19个）、A（自回归动量，每币种4个）。E、S、A类为币种特定；P类包含三种稳定币脱锚信号，对每个币种模型均纳入；F类取自FRED并按发布滞后对齐。

表 A.11: Complete Variable Definitions and Formulas (104 Variables)

ID	Variable Name	Symbol	Transform	Formula & Explanation	Expected Sign
Panel M. Macroeconomic & External Market (12 variables)					
M1	3-Month Treasury Yield (change)	`us3m_treasury_yield`	1st diff	$\Delta r_t^{3M} = r_t^{3M} - r_{t-1}^{3M}$; short-term risk-free rate change	$+$ (hike $\rightarrow$ reserve yield $\uparrow$ $\rightarrow$ minting $\uparrow$)
M2	SOFR–RRP Spread (change)	`sofr_rrp_spread`	1st diff	Change in spread between SOFR and Reverse Repo rate	$+$ (tighter funding $\rightarrow$ stablecoin demand $\uparrow$)
M3	Global Policy Risk Index (change)	`gprd_index`	1st diff	News-based policy-uncertainty index change	$-$ (regulatory threat $\rightarrow$ redemptions)
M4	Economic Policy Uncertainty Index (change)	`epu_index`	1st diff	Baker–Bloom–Davis EPU change	$-$ (uncertainty $\rightarrow$ flight to safety)
M5	S&P 500 Index (log return)	`sp500_index`	Log-diff	$\Delta \ln P_t^{SP500}$; proxy for global risk appetite	$+$ (risk-on $\rightarrow$ crypto inflows)
M6	Gold Price (log return)	`gold_price`	Log-diff	$\Delta \ln P_t^{Gold}$; safe-haven asset return	Ambiguous
M7	Bitcoin Price (log return)	`bitcoin_price`	Log-diff	$\Delta \ln P_t^{BTC}$; crypto market beta	$+$ (crypto rally $\rightarrow$ stablecoin demand)
M8	US Dollar Index (log return)	`dxy`	Log-diff	$\Delta \ln DXY_t$; dollar strength	$-$ (stronger dollar $\rightarrow$ dollar-token demand ambiguous)
M9	USDCNH Exchange Rate (log return)	`usdcnh_exchange_rate`	Log-diff	$\Delta \ln S_t^{USDCNH}$; offshore RMB/dollar	$+$ (CNH depreciation $\rightarrow$ USDT premium)
M10	3-Month Treasury Yield (level)	`us3m_treasury_yield_level`	Level	r_t^{3M}; level of short-term yield	$\pm$
M11	SOFR–RRP Spread (level)	`sofr_rrp_spread_level`	Level	Level of SOFR–RRP spread	$\pm$
M12	Global Policy Risk Index (level)	`gprd_index_level`	Level	Level of GPRD index	$-$
Panel F. Fed Liquidity Plumbing (8 variables; FRED, publication-lag aligned)					
F1	SOFR (change)	`sofr`	1st diff	ΔSecured Overnight Financing Rate; daily, lagged 1 day	$-$ (tighter funding $\rightarrow$ risk-off)
F2	SOFR (level)	`sofr_level`	Level	Level of SOFR	$\pm$
F3	ON RRP Volume (change)	`rrp_volume`	1st diff	Δovernight reverse-repo take-up; daily, lagged 1 day	$-$ (parked cash $\rightarrow$ less dry powder)
F4	ON RRP Volume (level)	`rrp_volume_level`	Level	Level of ON RRP volume	$-$
F5	TGA Balance (change)	`tga_balance`	1st diff	ΔTreasury General Account balance; daily, lagged 1 day	$-$ (TGA rebuild drains reserves)
F6	TGA Balance (level)	`tga_balance_level`	Level	Level of TGA balance	$-$
F7	Net Liquidity (change)	`net_liquidity`	1st diff	Δ(Fed assets $-$ TGA $-$ ON RRP); weekly assets lagged 8 days	$+$ (liquidity injection $\rightarrow$ risk appetite)
F8	Net Liquidity (level)	`net_liquidity_level`	Level	Level of net liquidity	$+$
Panel G. Crypto Global Indicators (19 variables)					
G1	Total Market Cap (all coins)	`total_market_cap_all_coins`	Log-diff	$\Delta \ln \mathrm{Cap}_t^{all}$; aggregate crypto market size	$+$

Table A.11 continued

ID	Variable Name	Symbol	Transform	Formula & Explanation	Expected Sign
G2	Large-Cap Market Cap	`total_market_cap_large_cap`	Log-diff	$\Delta \ln \text{Cap}_t^{large}$	$+$
G3	Mid-Cap Market Cap	`total_market_cap_mid_cap`	Log-diff	$\Delta \ln \text{Cap}_t^{mid}$	$+$
G4	Small-Cap Market Cap	`total_market_cap_small_cap`	Log-diff	$\Delta \ln \text{Cap}_t^{small}$	$+$
G5	Market Cap Index (all coins)	`total_market_cap_index_all_coins`	Log-diff	Index-version aggregate market cap change	$+$
G6	Total Open Interest (all coins)	`total_open_interest_all_coins`	Log-diff	$\Delta \ln \text{OI}_t^{all}$; aggregate leverage demand	$+$
G7	Large-Cap OI	`total_open_interest_large_cap`	Log-diff	Large-cap OI change	$+$
G8	Mid-Cap OI	`total_open_interest_mid_cap`	Log-diff	Mid-cap OI change	$+$
G9	Small-Cap OI	`total_open_interest_small_cap`	Log-diff	Small-cap OI change	$+$
G10	OI Index (all coins)	`total_open_interest_index_all_coins`	Log-diff	Index-version aggregate OI change	$+$
G11	Total Realized Cap (all coins)	`total_realized_cap_all_coins`	Log-diff	$\Delta \ln \text{RC}_t^{all}$; cost-basis weighted valuation	$+$
G12	Large-Cap Realized Cap	`total_realized_cap_large_cap`	Log-diff	Large-cap realized-cap change	$+$
G13	Mid-Cap Realized Cap	`total_realized_cap_mid_cap`	Log-diff	Mid-cap realized-cap change	$+$
G14	Small-Cap Realized Cap	`total_realized_cap_small_cap`	Log-diff	Small-cap realized-cap change	$+$
G15	Realized Cap Index	`total_realized_cap_index_all_coins`	Log-diff	Index-version realized-cap change	$+$
G16	DeFi Total Value Locked	`defi_tvl`	Log-diff	$\Delta \ln \text{TVL}_t$; DeFi ecosystem size	$+$ (DeFi growth $\rightarrow$ stablecoin lock-up)
G17	Fear & Greed Index	`fear_greed_index`	Level	Alternative.me sentiment index (0–100)	$+$ (greed $\rightarrow$ inflows)
G18	Spot Volume (all assets)	`spot_volume_all_assets`	Log-diff	Aggregate spot trading volume change	$+$
G19	USDT OTC Premium	`usdt_otc_premium`	Level	Offshore CNH premium of USDT over USD	$+$ (premium $\rightarrow$ minting arbitrage)
Panel L. BTC Derivatives Leverage (28 variables)					
L1	Perpetual Funding Rate	`futures_perpetual_funding_rate`	Level	OI-weighted funding rate; > 0 means longs pay shorts	$+$
L2	Perpetual Funding Rate V2	`futures_perpetual_funding_rate_v2`	Level	Liquidity-weighted refined funding rate	$+$
L3	Futures Annualized Rolling Basis (3M)	`futures_annualized_rolling_basis_3m`	Level	$\frac{1}{\tau}(F_{t,t+\tau}/S_t - 1)$; cash-and-carry incentive	$+$

Table A.11 continued

ID	Variable Name	Symbol	Transform	Formula & Explanation	Expected Sign
L4	Perpetuals Reference Rate	`perpetuals_reference_rate`	Level	Reference funding rate across exchanges	+
L5	Futures Open Interest	`futures_open_interest`	Log-diff	Total BTC futures OI change	+
L6	Perpetual OI	`futures_open_interest_perpetual`	Log-diff	Perpetual-only OI change	+
L7	Cash-Margined Futures OI	`cash_margined_futures_open_interest`	Log-diff	Stablecoin-margined futures OI change	+ (direct stablecoin demand)
L8	Crypto-Margined Futures OI	`crypto_margined_futures_open_interest`	Log-diff	BTC/ETH-margined futures OI change	Structural signal
L9	Perpetual Cash-Margin OI	`futures_open_interest_cash_margin_perpetual`	Log-diff	Stablecoin-margined perpetual OI change	+
L10	Perpetual Crypto-Margin OI	`futures_open_interest_crypto_margin_perpetual`	Log-diff	BTC/ETH-margined perpetual OI change	Structural signal
L11	Futures Estimated Leverage Ratio	`futures_estimated_leverage_ratio`	Level	$\text{OI}_t/\text{Exchange Balance}_t$	+ (leverage up → margin demand)
L12	Total Futures Liquidations	`futures_total_liquidations_total`	Log-diff	Total liquidation volume change	− (liquidations → redemptions)
L13	Long Liquidations	`futures_long_liquidations_total`	Log-diff	Long liquidation volume change	−
L14	Short Liquidations	`futures_short_liquidations_total`	Log-diff	Short liquidation volume change	+ (short squeeze → rally)
L15	Long-Liquidation Dominance	`futures_long_liquidations_dominance`	Level	Long liquidations / total liquidations	−
L16	Futures Volume	`futures_volume`	Log-diff	Total futures trading volume change	+
L17	Perpetual Futures Volume	`futures_volume_perpetual`	Log-diff	Perpetual futures volume change	+
L18	Futures Buy Volume (24h)	`futures_buy_volume_24h`	Log-diff	24h buy volume change	+
L19	Futures Sell Volume (24h)	`futures_sell_volume_24h`	Log-diff	24h sell volume change	−
L20	Perpetual Buy Volume (24h)	`futures_buy_volume_perpetual_24h`	Log-diff	Perpetual buy volume change	+
L21	Perpetual Sell Volume (24h)	`futures_sell_volume_perpetual_24h`	Log-diff	Perpetual sell volume change	−
L22	Perpetual Volume Delta	`futures_volume_delta_perpetual`	Level	Buy–sell volume difference in perpetuals	+
L23	Options Volume (24h)	`options_volume_24h`	Log-diff	24h options volume change	+
L24	Options Volume Put/Call Ratio	`options_volume_put_call_ratio`	Level	Put volume / call volume	Contrarian
L25	Options OI Put/Call Ratio	`options_open_interest_put_call_ratio`	Level	Put OI / call OI	Contrarian
L26	Realized Volatility (1 month)	`realized_volatility_1_month`	Level	1-month realized volatility	+ (vol → safe-haven demand)
L27	Position Openings	`position_openings`	Log-diff	Number of new positions opened	+

Table A.11 continued

ID	Variable Name	Symbol	Transform	Formula & Explanation	Expected Sign
L28	Position Closures	`position_closures`	Log-diff	Number of positions closed	−
Panel E. Exchange Flows (11 variables, coin-specific)					
E1	Exchange Inflow Volume	`exchange_inflow_volume_total`	Log-diff	Total inflow to exchanges	+ (margin deployment)
E2	Exchange Outflow Volume	`exchange_outflow_volume_total`	Log-diff	Total outflow from exchanges	+ (withdraw-and-hoard)
E3	Exchange Balance Total	`exchange_balance_total`	Log-diff	Total exchange balance change	+ (available margin)
E4	Depositing Addresses	`depositing_addresses`	Log-diff	Number of addresses depositing	+ (user breadth)
E5	Withdrawing Addresses	`withdrawing_addresses`	Log-diff	Number of addresses withdrawing	+ (user breadth)
E6	Exchange Reliance Ratio	`exchange_reliance_ratio`	Level	Exchange balance / total supply	Structural signal
E7	Exchange Reshuffling Ratio	`exchange_reshuffling_ratio`	Level	Internal reallocation ratio	Synchronous signal
E8	Aggregated Exchange Reliance	`exchange_aggregated_reliance_ratio`	Level	Aggregate exchange reliance across major venues	Structural signal
E9	Aggregated Reshuffling Ratio	`exchange_aggregated_reshuffling_ratio`	Level	Aggregate reallocation ratio	Synchronous signal
E10	In-House Exchange Volume	`in_house_exchange_volume`	Log-diff	Internal exchange transfer volume change	Synchronous signal
E11	Inter-Exchange Volume	`inter_exchange_volume`	Log-diff	Cross-exchange transfer volume change	Leading signal
Panel P. Depeg & Arbitrage (3 variables)					
P1	USDT Depeg	`usdt_depeg`	Level	$\lvert P_t^{USDT} - 1\rvert$ or deviation from \$1 peg	+ (arbitrage minting/redeeming)
P2	USDC Depeg	`usdc_depeg`	Level	USDC price deviation from \$1	+ (arbitrage flows)
P3	DAI Depeg	`dai_depeg`	Level	DAI price deviation from \$1	+ (arbitrage flows)
Panel S. On-chain Structure (19 variables, coin-specific)					
S1	Total Addresses	`total_addresses_aggregated`	Log-diff	Total address count change	+ (user growth)
S2	New Addresses	`new_addresses_aggregated`	Log-diff	New address count change	+ (new demand)
S3	Receiving Addresses	`receiving_addresses_aggregated`	Log-diff	Receiving address count change	+
S4	Sending Addresses	`sending_addresses_aggregated`	Log-diff	Sending address count change	+
S5	Active Addresses	`active_addresses_aggregated`	Log-diff	Active address count change	+ (usage intensity)
S6	Transfer Count	`transfer_count_aggregated`	Log-diff	Number of transfers change	+
S7	Transfer Rate	`transfer_rate_aggregated`	Log-diff	Transfer frequency change	+
S8	Transfer Volume Total	`transfer_volume_total_aggregated`	Log-diff	Total on-chain transfer volume change	+

Table A.11 continued

ID	Variable Name	Symbol	Transform	Formula & Explanation	Expected Sign
S9	Mean Transfer Volume	`transfer_volume_mean_aggregated`	Log-diff	Average transfer size change	+
S10	Coin Days Destroyed	`coin_days_destroyed_cdd`	Level	$\sum \text{CoinsMoved}_i \cdot \text{DaysHeld}_i$	− (distribution → selling)
S11	Dormancy	`dormancy`	Level	CDD / transaction volume	Structural signal
S12	NVT Ratio	`nvt_ratio`	Level	Market cap / on-chain volume	Valuation signal
S13	Hodler Net Position Change	`hodler_net_position_change`	Level	Net position change of long-term holders	+ (accumulation → minting)
S14	Liveliness	`liveliness`	Level	Cumulative CDD / total coin-days	Structural signal
S15	Activity Retention Rate	`activity_retention_rate`	Level	Retention of active users	+
S16	Holder Accumulation Ratio	`holder_accumulation_ratio`	Level	Share of supply in accumulation addresses	+
S17	Holder Retention Rate	`holder_retention_rate`	Level	Retention of existing holders	+
S18	Gini Coefficient	`gini_coefficient`	Level	Address-holding concentration	Regime signal
S19	Herfindahl Index	`herfindahl_index`	Level	$\sum w_j^2$ of holding shares	Regime signal
Panel A. Autoregressive Momentum (4 variables, coin-specific)					
A1	Issuance Lag 1	`l1_sup_ret_coin`	Level	$y_{t-1} = \Delta \ln S_{t-1}$	+/ (momentum/mean reversion)
A2	Issuance Lag 2	`l2_sup_ret_coin`	Level	y_{t-2}	Data-dependent
A3	4-Week Moving Average	`ma4_sup_ret_coin`	Level	$\frac{1}{4}\sum_{k=0}^{3} y_{t-k}$	+ (trend)
A4	Issuance Volatility (4W)	`sup_vol_coin`	Level	4-week rolling standard deviation of y_t	− (high uncertainty)

Notes: All stock-type variables enter as log-differences; rates, indices, and ratios enter as levels with within-training-set Min-Max normalization; all variables are median-imputed and clipped to $[0,1]$ after normalization. The E, S, and A categories are coin-specific; the P category includes USDT, USDC, and DAI depeg signals for every coin model; the F category is drawn from FRED with publication-lag alignment. Total feature pool per coin: $12+8+19+28+11+3+19+4=104$.

表 A.12: 完整变量定义与公式（104个变量）

编号	变量名	缩写	形式	公式与说明	预期方向
面板M：宏观与外部市场（12个变量）					
M1	3个月美债收益率（变化）	`us3m_treasury_yield`	一阶差分	$\Delta r_t^{3M} = r_t^{3M} - r_{t-1}^{3M}$；短期无风险利率变化	+（加息→储备收益↑→铸币↑）
M2	SOFR–RRP利差（变化）	`sofr_rrp_spread`	一阶差分	SOFR与逆回购利率利差的变化	+（资金趋紧→稳定币需求↑）
M3	全球政策风险指数（变化）	`gprd_index`	一阶差分	基于新闻的政策不确定性指数变化	（监管威胁→赎回）
M4	经济政策不确定性指数（变化）	`epu_index`	一阶差分	Baker–Bloom–Davis EPU变化	（不确定性→避险）
M5	标普500指数（对数收益）	`sp500_index`	对数差分	$\Delta \ln P_t^{SP500}$；全球风险偏好代理	+（风险情绪↑→加密流入）
M6	黄金价格（对数收益）	`gold_price`	对数差分	$\Delta \ln P_t^{Gold}$；避险资产收益	不确定
M7	比特币价格（对数收益）	`bitcoin_price`	对数差分	$\Delta \ln P_t^{BTC}$；加密市场贝塔	+（加密上涨→稳定币需求）
M8	美元指数（对数收益）	`dxy`	对数差分	$\Delta \ln DXY_t$；美元强弱	±
M9	离岸人民币汇率（对数收益）	`usdcnh_exchange_rate`	对数差分	$\Delta \ln S_t^{USDCNH}$；离岸人民币/美元	+（人民币贬值→USDT溢价）
M10	3个月美债收益率（原值）	`us3m_treasury_yield_level`	原值	r_t^{3M}；短期收益率水平	±
M11	SOFR–RRP利差（原值）	`sofr_rrp_spread_level`	原值	SOFR–RRP利差水平	±
M12	全球政策风险指数（原值）	`gprd_index_level`	原值	GPRD指数水平	
面板F：美联储流动性管道（8个变量；FRED，按发布滞后对齐）					
F1	SOFR（变化）	`sofr`	一阶差分	担保隔夜融资利率变化；日度，滞后1天	（资金趋紧→避险）
F2	SOFR（原值）	`sofr_level`	原值	SOFR水平	±
F3	隔夜逆回购规模（变化）	`rrp_volume`	一阶差分	ON RRP规模变化；日度，滞后1天	（资金停泊→可用弹药减少）
F4	隔夜逆回购规模（原值）	`rrp_volume_level`	原值	ON RRP规模水平	
F5	财政部一般账户余额（变化）	`tga_balance`	一阶差分	TGA余额变化；日度，滞后1天	（TGA重建抽走准备金）
F6	财政部一般账户余额（原值）	`tga_balance_level`	原值	TGA余额水平	
F7	净流动性（变化）	`net_liquidity`	一阶差分	Δ（美联储资产−TGA−ON RRP）；周度资产滞后8天	+（流动性注入→风险偏好↑）
F8	净流动性（原值）	`net_liquidity_level`	原值	净流动性水平	+
面板G：加密全局指标（19个变量）					
G1	全币种总市值	`total_market_cap_all_coins`	对数差分	$\Delta \ln \mathrm{Cap}_t^{all}$；加密市场总规模	+
G2	大市值币总市值	`total_market_cap_large_cap`	对数差分	大市值币总市值变化	+
G3	中市值币总市值	`total_market_cap_mid_cap`	对数差分	中市值币总市值变化	+
G4	小市值币总市值	`total_market_cap_small_cap`	对数差分	小市值币总市值变化	+
G5	全币种总市值指数	`total_market_cap_index_all_coins`	对数差分	指数版总市值变化	+
G6	全币种总未平仓量	`total_open_interest_all_coins`	对数差分	$\Delta \ln \mathrm{OI}_t^{all}$；总杠杆需求	+

表 A.12 续

编号	变量名	缩写	形式	公式与说明	预期方向
G7	大市值币OI	total_open_interest_large_cap	对数差分	大市值币OI变化	+
G8	中市值币OI	total_open_interest_mid_cap	对数差分	中市值币OI变化	+
G9	小市值币OI	total_open_interest_small_cap	对数差分	小市值币OI变化	+
G10	全币种OI指数	total_open_interest_index_all_coins	对数差分	指数版总OI变化	+
G11	全币种已实现市值	total_realized_cap_all_coins	对数差分	$\Delta \ln \mathrm{RC}_t^{all}$；成本基础加权估值	+
G12	大市值币已实现市值	total_realized_cap_large_cap	对数差分	大市值币已实现市值变化	+
G13	中市值币已实现市值	total_realized_cap_mid_cap	对数差分	中市值币已实现市值变化	+
G14	小市值币已实现市值	total_realized_cap_small_cap	对数差分	小市值币已实现市值变化	+
G15	已实现市值指数	total_realized_cap_index_all_coins	对数差分	指数版已实现市值变化	+
G16	DeFi总锁仓价值	defi_tvl	对数差分	$\Delta \ln \mathrm{TVL}_t$；DeFi生态规模	+（DeFi增长→稳定币锁定）
G17	恐惧与贪婪指数	fear_greed_index	原值	Alternative.me情绪指数（0–100）	+（贪婪→流入）
G18	全资产现货成交量	spot_volume_all_assets	对数差分	总现货交易量变化	+
G19	USDT OTC溢价	usdt_otc_premium	原值	USDT离岸人民币溢价	+（溢价→铸币套利）
面板L：BTC衍生品杠杆（28个变量）					
L1	永续资金费率	futures_perpetual_funding_rate	原值	OI加权资金费率；>0表示多头付空头	+
L2	永续资金费率V2	futures_perpetual_funding_rate_v2	原值	流动性加权优化版资金费率	+
L3	期货年化滚动基差（3M）	futures_annualized_rolling_basis_3m	原值	$\frac{1}{\tau}(F/S-1)$；期现套利激励	+
L4	永续参考利率	perpetuals_reference_rate	原值	各交易所参考资金费率	+
L5	期货未平仓量	futures_open_interest	对数差分	BTC期货总OI变化	+
L6	永续OI	futures_open_interest_perpetual	对数差分	仅永续合约OI变化	+
L7	现金保证金期货OI	cash_margined_futures_open_interest	对数差分	稳定币保证金期货OI变化	+（直接稳定币需求）
L8	加密保证金期货OI	crypto_margined_futures_open_interest	对数差分	BTC/ETH保证金期货OI变化	结构信号
L9	永续现金保证金OI	futures_open_interest_cash_margin_perpetual	对数差分	稳定币保证金永续OI变化	+

表 A.12 续

编号	变量名	缩写	形式	公式与说明	预期方向
L10	永续加密保证金OI	`futures_open_interest_crypto_margin_perpetual`	对数差分	BTC/ETH保证金永续OI变化	结构信号
L11	期货估算杠杆率	`futures_estimated_leverage_ratio`	原值	OI_t/交易所余额$_t$	+（杠杆↑→保证金需求）
L12	期货总清算量	`futures_total_liquidations_total`	对数差分	总清算量变化	（清算→赎回）
L13	多头清算量	`futures_long_liquidations_total`	对数差分	多头清算量变化	
L14	空头清算量	`futures_short_liquidations_total`	对数差分	空头清算量变化	+（空头挤压→上涨）
L15	多头清算主导度	`futures_long_liquidations_dominance`	原值	多头清算/总清算	
L16	期货成交量	`futures_volume`	对数差分	期货总成交量变化	+
L17	永续期货成交量	`futures_volume_perpetual`	对数差分	永续期货成交量变化	+
L18	期货买入量（24h）	`futures_buy_volume_24h`	对数差分	24h买入量变化	+
L19	期货卖出量（24h）	`futures_sell_volume_24h`	对数差分	24h卖出量变化	
L20	永续买入量（24h）	`futures_buy_volume_perpetual_24h`	对数差分	永续买入量变化	+
L21	永续卖出量（24h）	`futures_sell_volume_perpetual_24h`	对数差分	永续卖出量变化	
L22	永续成交量Delta	`futures_volume_delta_perpetual`	原值	永续买卖量差	+
L23	期权成交量（24h）	`options_volume_24h`	对数差分	24h期权成交量变化	+
L24	期权成交量P/C比	`options_volume_put_call_ratio`	原值	看跌/看涨成交量比	反向
L25	期权持仓P/C比	`options_open_interest_put_call_ratio`	原值	看跌/看涨持仓量比	反向
L26	一个月已实现波动率	`realized_volatility_1_month`	原值	一个月已实现波动率	+（波动↑→避险需求）
L27	新开仓量	`position_openings`	对数差分	新开仓头寸数	+
L28	平仓量	`position_closures`	对数差分	平仓头寸数	
面板E：交易所资金流（11个变量，币种特定）					
E1	交易所流入总量	`exchange_inflow_volume_total`	对数差分	交易所总流入量变化	+（保证金部署）
E2	交易所流出总量	`exchange_outflow_volume_total`	对数差分	交易所总流出量变化	+（提现囤积）
E3	交易所总余额	`exchange_balance_total`	对数差分	交易所总余额变化	+（可用保证金）
E4	充值地址数	`depositing_addresses`	对数差分	充值地址数变化	+（用户广度）
E5	提现地址数	`withdrawing_addresses`	对数差分	提现地址数变化	+（用户广度）
E6	交易所依赖比率	`exchange_reliance_ratio`	原值	交易所余额/总供应量	结构信号

表 A.12 续

编号	变量名	缩写	形式	公式与说明	预期方向
E7	交易所再配置比率	exchange_reshuffling_ratio	原值	内部再配置比率	同步信号
E8	聚合交易所依赖比率	exchange_aggregated_reliance_ratio	原值	主要交易所聚合依赖比率	结构信号
E9	聚合再配置比率	exchange_aggregated_reshuffling_ratio	原值	聚合再配置比率	同步信号
E10	交易所内部转账量	in_house_exchange_volume	对数差分	交易所内部转账量变化	同步信号
E11	交易所间转账量	inter_exchange_volume	对数差分	跨所转账量变化	领先信号
面板P：脱锚与套利（3个变量）					
P1	USDT脱锚	usdt_depeg	原值	USDT价格偏离1美元幅度	+（套利铸币/赎回）
P2	USDC脱锚	usdc_depeg	原值	USDC价格偏离1美元幅度	+（套利流动）
P3	DAI脱锚	dai_depeg	原值	DAI价格偏离1美元幅度	+（套利流动）
面板S：链上结构（19个变量，币种特定）					
S1	总地址数	total_addresses_aggregated	对数差分	总地址数变化	+（用户增长）
S2	新增地址数	new_addresses_aggregated	对数差分	新增地址数变化	+（新需求）
S3	接收地址数	receiving_addresses_aggregated	对数差分	接收地址数变化	+
S4	发送地址数	sending_addresses_aggregated	对数差分	发送地址数变化	+
S5	活跃地址数	active_addresses_aggregated	对数差分	活跃地址数变化	+（使用强度）
S6	转账笔数	transfer_count_aggregated	对数差分	转账笔数变化	+
S7	转账频率	transfer_rate_aggregated	对数差分	转账频率变化	+
S8	转账总量	transfer_volume_total_aggregated	对数差分	链上转账总量变化	+
S9	平均转账量	transfer_volume_mean_aggregated	对数差分	平均转账规模变化	+
S10	币天销毁	coin_days_destroyed_cdd	原值	$\sum$ 转移币数$_i$ · 持币天数$_i$	（分配→抛售）
S11	休眠度	dormancy	原值	CDD/交易量	结构信号
S12	NVT比率	nvt_ratio	原值	市值/链上交易量	估值信号
S13	持有者净头寸变化	hodler_net_position_change	原值	长期持有者净头寸变化	+（积累→铸币）
S14	活跃度	liveliness	原值	累计CDD/总币天	结构信号
S15	活跃留存率	activity_retention_rate	原值	活跃用户留存率	+
S16	持有者积累比率	holder_accumulation_ratio	原值	积累地址持仓占比	+
S17	持有者留存率	holder_retention_rate	原值	现有持有者留存率	+
S18	基尼系数	gini_coefficient	原值	地址持有集中度	体制信号

表 A.12 续

编号	变量名	缩写	形式	公式与说明	预期方向
S19	赫芬达尔指数	herfindahl_index	原值	持仓份额的$\sum w_j^2$	体制信号
面板A：自回归动量（4个变量，币种特定）					
A1	发行量滞后1期	l1_sup_ret_coin	原值	$y_{t-1} = \Delta \ln S_{t-1}$	+/ （动量/均值回复）
A2	发行量滞后2期	l2_sup_ret_coin	原值	y_{t-2}	视数据而定
A3	4周移动平均	ma4_sup_ret_coin	原值	$\frac{1}{4}\sum_{k=0}^{3} y_{t-k}$	+（趋势）
A4	发行量波动率（4周）	sup_vol_coin	原值	y_t的4周滚动标准差	（高不确定性）

注：所有存量变量以对数差分进入；利率、指数、比率类以原值进入，在训练集内做Min-Max归一化；所有变量经中位数填充后裁剪到$[0, 1]$。E、S、A类为币种特定；P类包含三种稳定币脱锚信号，对每个币种模型均纳入；F类取自FRED并按发布滞后对齐。每币种总特征数：$12 + 8 + 19 + 28 + 11 + 3 + 19 + 4 = 104$。

附录 B. Validation Evidence for Panel–Channel Selection / 面板–通道选择的验证集证据

Table B.13 reports, for each forecast horizon, the validation walk-forward MAE of all five candidate averaging channels evaluated by the combination layer: three-strategy or four-stack averages built on the 96-variable base panel, the 104-variable liquidity-augmented panel, or both. Channel selection uses the inner-validation sample only; the blind test set is evaluated exactly once. The final design adopts the validation winner at $h = 1$ (avg4, 104-var), $h = 7$, and $h = 30$ (avg3, 96-var). At $h = 90$ the validation winner is the 104-var three-strategy average, but its larger forecast amplitudes do not survive the low-drift 2026 test fold (e.g., USDT fold 4: validation R^2_{OS} +0.60, test R^2_{OS} −528%), so the 96-var four-stack channel is retained under a cross-regime robustness prior that favours configurations surviving regime rotation. The liquidity-augmented panel is therefore used exactly where validation evidence supports it, and its information reaches the longer horizons through the ex-ante feature-selection stage rather than through unregularized amplitude.

表 B.13 报告了每个预测期限下组合层评估的全部五个候选平均通道的验证集前向 MAE：基于 96 变量基准面板、104 变量流动性增强面板或两者的三策略/四栈平均。通道选择仅使用内验证样本；盲测集只评估一次。最终设计在 $h = 1$（avg4，104 变量）、$h = 7$ 与 $h = 30$（avg3，96 变量）采纳验证集最优通道。在 $h = 90$，验证集最优为 104 变量三策略平均，但其更大的预测振幅无法在低漂移的 2026 年测试折中存活（如 USDT 第 4 折：验证 R^2_{OS} 为 +0.60，盲测 R^2_{OS} 为 −528%），因此在跨体制稳健先验下保留 96 变量四栈通道——该先验偏好能在体制轮换中存活的配置。流动性增强面板因而恰好用于验证证据支持之处，其信息经由事前特征选择阶段传导至更长期限，而非经由未正则化的振幅。

表 B.13: Validation walk-forward MAE of the five candidate averaging channels at each horizon. **Bold**: validation minimum; †: channel adopted in the final design. At $h = 90$ the validation winner (avg3, 104-var panel) is rejected under a cross-regime robustness prior—its larger forecast amplitudes collapse in the low-drift 2026 test fold—and the 96-var four-stack channel is retained instead.

Channel	$h = 1$	$h = 7$	$h = 30$	$h = 90$
avg3, 96-var panel	0.5056	**0.5775**†	**0.7780**†	1.1832
avg4, 96-var panel	0.5045	0.5810	0.7812	1.1667†
avg3, 104-var panel	0.5017	0.5899	0.7835	**1.1224**
avg4, 104-var panel	**0.5017**†	0.5888	0.7843	1.1451
avg8, both panels	0.5027	0.5857	0.7803	1.1519